

Security-based fuzzy control for USV via a memory-dependent event-triggered mechanism under hybrid cyber-attacks

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ABSTRACT

This paper addresses the event-triggered security control problem for unmanned surface vehicle (USV) under hybrid cyber-attacks. A closed-loop switched system model is established using the T-S fuzzy framework to represent USV subjected to both hybrid cyber-attacks and external disturbances. Then, a memory-dependent event-triggered mechanism (MDETm) is introduced, which dynamically adapts the triggered threshold according to historical attack intensity and fuzzy membership information. This mechanism not only saves limited communication resources but also effectively mitigates the adverse impacts of state information distortion caused by attacks, thereby enhancing control performance. Furthermore, a novel Lyapunov-Krasovskii functional (LKF) is constructed via a two-side looped-functional method (TSLFM). This approach makes full use of state information over sampling intervals and incorporates fuzzy membership knowledge, leading to reduced design conservatism. Based on this, sufficient conditions are derived to guarantee exponential mean-square stability (EMSS) of the USV system with a prescribed H_∞ performance level. Finally, several comparative simulation results are presented to demonstrate the effectiveness and superiority of the proposed approach.

1. Introduction

Unmanned surface vehicle (USV), characterized by good maneuverability, strong adaptability, high autonomy, and low manufacturing cost, has emerged as an important autonomous marine platform and is widely used in various maritime applications (Wang et al. (2024); Yin et al. (2025b)). However, as the technology continues to evolve, it also introduces a range of challenges. These challenges, which have consequently drawn substantial attention from both academia and industry, sparking intensive research aimed at addressing these problem (Wang and Li (2025); Yan et al. (2025)).

Among these challenges, the foremost one is the highly nonlinear and uncertain dynamic model of USV, which results from the complex and ever-changing hydrodynamic environment, dynamic operating conditions, and significant external disturbances (Ning et al. (2023)). Faced with such an intricate system, conventional linear control strategies often struggle to achieve satisfactory performance. As a viable alternative, the Takagi-Sugeno (T-S) fuzzy model (Takagi and Sugeno (1985); Li et al. (2013)) has been widely recognized as an effective methodology for approximating nonlinear systems. Consequently, in recent years, numerous control methods based on the T-S fuzzy model have been proposed for USV systems (Zheng et al. (2023); Wang et al. (2025b)). Specifically, to address USVs with unmeasured velocities, Wang et al. (2018) developed a network-based fuzzy output-feedback controller for dynamic positioning. A fuzzy reliable tracking control scheme was proposed for USV in Zheng et al. (2023), which ensures square-exponential stability and convergence of the tracking error to zero.

Moreover, Wang et al. (2025b) presented an observer-based fuzzy sliding mode control strategy and ensures the finite-time stability of USV.

Furthermore, with the rapid development of wireless communication technology, network control systems (NCSs) have become a prominent solution in USV, offering benefits like remote operation and easy scalability compared to conventional methods (Wang et al. (2023); Xu et al. (2024); Cheng et al. (2025)). However, this reliance on open communication networks also exposes them to cybersecurity threats, presenting another critical challenge. Among these, Denial-of-Service (DoS) attacks are particularly prevalent, in which adversaries overload the network with excessive traffic, leading to system paralysis (Ye et al. (2023)). Equally concerning are deception attacks, in which adversaries forge or tamper with critical data such as location and speed measurements Wu et al. (2023a). Such falsified data can mislead the navigation system of the USV, ultimately compromising its operational safety. To mitigate these risks, numerous security control strategies have recently been proposed for USV, including defenses against DoS attacks (Wang et al. (2025a); Ma et al. (2024); Song et al. (2025)) and countermeasures for deception attacks (Jiang et al. (2024); Sun et al. (2025)). Wang et al. (2025a) investigates the fault-tolerant formation control problem under DoS attacks and provides a quantitative analysis of the relationships between attack frequency, sampling interval and system performance. Additionally, an observer-based methods have been developed to estimate system states and reconstruct the deception attack signals, enabling precise compensation and ensuring security (Sun et al. (2025)). However, most existing studies mainly focus on single cyber-attacks, while the impact of hybrid cyber-attacks has not been sufficiently investigated. This limitation may lead to an

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incomplete assessment of security risks and degraded control performance in practical networked USV systems. Therefore, how to further investigate and address these issues becomes one of the main motivations of this paper.

Another challenge in USV NCSs is limited communication bandwidth. Conventional controllers, such as those developed in Hao et al. (2021); Zheng et al. (2023), are updated at each fixed sampling interval, resulting in excessive bandwidth occupation and increased computational burden (Du et al. (2025)). To address this, event-triggered mechanisms (ETMs) transmit control signals only when a predefined condition is met, reducing unnecessary communication while maintaining performance. Consequently, various event-triggered control methods have been proposed (Ma et al. (2020); Yin et al. (2025a)). However, these schemes focus solely on the error between the current and the most recently triggered data. Under significant system fluctuations, this error can become negligible at the peaks or troughs of the response curve, leading the ETM to discard useful data packets (Xu et al. (2025); Liu et al. (2025)). To address this, memory-based ETM (METM) incorporates historical triggered data into the triggering condition, which can be further developed into an adaptive scheme to dynamically adjust the threshold, thereby optimizing data transmission and tracking performance (Dong et al. (2024), Gu et al. (2024)). Despite these advantages, the reduced communication inherent in ETMs inevitably amplifies the system's vulnerability to cyber-attacks, consequently degrading system stability. Consequently, securing ETM-based systems has garnered significant attention; for instance, Han et al. (2023) studied fuzzy resilient ETM dynamic positioning controllers for USVs. Nevertheless, commonly adopted Lyapunov–Krasovskii functional (LKF) for networked fuzzy systems (Ding and Zhu (2024); Luo et al. (2025)) rely heavily on state variations while neglecting intrinsic structural characteristics like fuzzy membership function (FMF), which may results in design conservatism (You et al. (2023)). Consequently, to simultaneously alleviate the communication burden and enhance control performance, this paper integrates FMF into the MDETm and LKF, while effectively accommodating imperfectly matched FMF by leveraging global membership boundary information during the stability analysis.

Based on the abovementioned analysis, this article will focus on the stability and stabilization problem for the T-S fuzzy USV systems under hybrid attacks by proposing a novel LKF and MDETm. The main contributions are as follows:

(1) An improved MDETm is proposed to conserve system resources and defend against hybrid cyber-attacks. By establishing a hybrid attack model, historical state information is utilized to mitigate the adverse impacts of data distortion and blockage. Meanwhile, the event-triggered threshold is adaptively adjusted based on historical attack conditions and trends, thereby enhancing the system's flexibility and adaptability to diverse attack scenarios.

(2) A unified framework is established to integrate the FMF into the MDETm, LKF and the corresponding

control synthesis. Specifically, distinct MDETms are designed according to the characteristics of different fuzzy subsystems. Accommodating hybrid cyber-attacks, network delays, and imperfectly matched FMF, a switched controller is developed to ensure the system's EMSS and a prescribed H_∞ performance.

(3) Based on the TSLFM, a novel relaxed fuzzy LKF is constructed to fully capture state information over $[x(t_k - t), x(t - \tau)]$ and $[x(t - \tau), x(t_{k+1} - \tau)]$. By removing the restriction that all Lyapunov matrices must be positive-definite and introducing fuzzy zero equations with free-weighting matrices, this approach significantly reduces design conservatism, yielding relaxed stability criteria and improved overall control performance.

Notation: $\mathbb{R}^n$ represents the n -dimensional Euclidean space, while $\mathbb{R}^{n \times m}$ is the set of real $n \times m$ matrices. I and 0 denote the identity matrix and zero matrix of appropriate dimensions. $\sup\{\cdot\}$ and $\inf\{\cdot\}$ refer to the supremum and infimum. The notation $\text{sym}(X)$ stands for $X + X^T$. The asterisk $*$ represents the elements induced by symmetry in a symmetric block matrix. $\text{diag}\{\dots\}$ denotes a block-diagonal matrix. $E\{\cdot\}$ indicates the mathematical expectation.

2. Problem formulation and preliminaries

2.1. Fuzzy modeling of USV

First, establish the dynamic model of USV. As illustrated in Fig. 1, the spatial motion of the USV is described utilizing two coordinate systems. Specifically, X_B , Y_B , and Z_B represent the longitudinal, transverse, and normal axes of the body-fixed frame, respectively, while X_E , Y_E , and Z_E designate the axes of the earth-fixed reference frame.

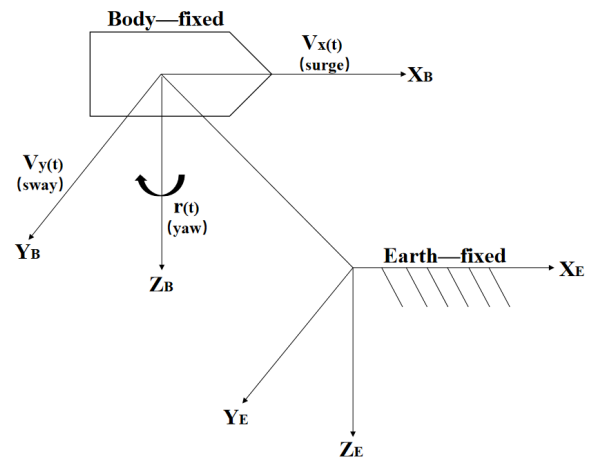


Figure 1: Body-fixed and earth-fixed reference frames of USV

The USV dynamic model is described as follows Wang et al. (2021):

$$M\dot{v}(t) + Nv(t) + G\eta(t) = u(t) + w(t). \quad (1)$$

where $\eta(t) = [p_x(t), p_y(t), \theta(t)]^T$ denotes the position and yaw angle of the USV, and $v(t) = [p(t), q(t), r(t)]^T$ represents its

surge, sway, and yaw velocities. $u(t) \in \mathbb{R}^3$ is the control input generated by the thrusters, and $w(t) \in \mathbb{R}^3$ is the bounded exogenous disturbance induced by waves, wind, and ocean currents. M , N , and G represent the inertia, damping, and mooring forces, respectively. Moreover, the following relationships hold:

$$\dot{\eta}(t) = \begin{bmatrix} \dot{p}_x(t) \\ \dot{p}_y(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p(t) \\ q(t) \\ r(t) \end{bmatrix} \quad (2)$$

$$= J(\theta)v(t)$$

where

$$J(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The model of equation (1) can be rewritten by:

$$\dot{v}(t) = A_\eta \eta(t) + A_v v(t) + B_{uw}[u(t) + w(t)] \quad (3)$$

where $A_\eta = -M^{-1}G$, $A_v = -M^{-1}N$, and $B_{uw} = M^{-1}$.

Define system states $x(t) = [\eta^T(t) v^T(t)]^T \in \mathbb{R}^6$. Then, rewrite the motion of the USV with disturbance by the following nonlinear model:

$$\dot{x}(t) = Ax(t) + Bu(t) + Bw(t) \quad (4)$$

with

$$A = \begin{bmatrix} 0 & J(\theta) \\ A_\eta & A_v \end{bmatrix}, B = \begin{bmatrix} 0 \\ B_{uw} \end{bmatrix}.$$

The T-S fuzzy models are described by the following r fuzzy rules, which can be used to approximate the above model:

Plant rule i : If $\theta_1(t)$ is $\mathfrak{N}_i^1$, $\theta_2(t)$ is $\mathfrak{N}_i^2$, and $\dots$ and $\theta_l(t)$ is $\mathfrak{N}_i^l$, then

$$\begin{cases} \dot{x}(t) = A_i x(t) + B_i u(t) + B_i w(t) \\ y(t) = C_i x(t) \end{cases} \quad (5)$$

where $i \in \{1, 2, \dots, r\}$, $y(t)$ is the measured output, and C_i are known output matrices.

The overall system can be derived from multiple fuzzy subsystems. Specifically,

$$\begin{cases} \dot{x}(t) = \sum_{i=1}^r g_i(\theta(t)) [A_i x(t) + B_i u(t) + D_i w(t)] \\ y(t) = \sum_{i=1}^r g_i(\theta(t)) C_i x(t) \end{cases} \quad (6)$$

Here, the fuzzy membership function (FMF) $g_i(\theta(t))$ is defined as $g_i(\theta(t)) = \rho_i(\theta(t)) / \sum_{i=1}^r \rho_i(\theta(t))$, where $\theta(t)$ denotes the premise variable vector, and the firing strength $\rho_i(\theta(t)) = \prod_{j=1}^l \mathfrak{N}_i^j(\theta_j(t))$. The FMF satisfies the following conditions:

$$g_i(\theta(t)) \geq 0, \sum_{i=1}^r g_i(\theta(t)) = 1, \sum_{i=1}^r \dot{g}_i(\theta(t)) = 0. \quad (7)$$

2.2. Hybrid cyber-attacks

Due to the exposed nature of wireless communication, networked USV systems are highly vulnerable to malicious intrusions. This paper investigates the impact of hybrid cyber-threats, specifically concurrent deception and DoS attacks.

Let $\eta(t) \in \{0, 1\}$ be a Bernoulli variable governing random deception attacks where the malicious signal $f(x(t))$ replaces legitimate measurements, with:

$$\mathbb{E}\{\eta(t)\} = \bar{\eta}, \quad \mathbb{E}\{(\eta(t) - \bar{\eta})^2\} = \bar{\eta}(1 - \bar{\eta}) \triangleq \rho^2 \quad (8)$$

where $\bar{\eta}$ and ρ^2 denote the expectation and variance of $\eta(t)$.

Assumption 1. (Wu et al. (2023b)): The deception attack signal $f(x(t))$ is constrained by a given constant matrix Φ :

$$\|f(x(t))\|_2 \leq \|\Phi x(t)\|_2 \quad (9)$$

Further, considering the DoS attacks, the actual data received by the controller is formulated as:

$$\tilde{x}(t) = (1 - \alpha(t)) [(1 - \eta(t))f(x(t)) + \eta(t)x(t)] \quad (10)$$

where $\alpha(t) \in \{0, 1\}$ dictates the DoS attack occurrence. For each attack cycle $n \in \mathbb{N}$, let H_n denote the end of the n -th active DoS phase, and μ_n be the duration of the subsequent dormant period, satisfying $\mu_n < H_{n+1} - H_n$. Accordingly, $\alpha(t) = 0$ indicates an available communication channel for $t \in [H_n, H_n + \mu_n)$, while $\alpha(t) = 1$ signifies a completely blocked channel during $t \in [H_n + \mu_n, H_{n+1})$.

Remark 1. It should be noted that the DoS attacks considered in this paper possess unknown attack periods. For any $n \in \mathbb{N}$, define $W_{1,n} \triangleq [H_n, H_n + \mu_n)$ and $W_{2,n} \triangleq [H_n + \mu_n, H_{n+1})$ as the n -th dormant and active phases of the DoS attack. Accordingly, the distinct cyber-attack scenarios are uniquely characterized by the combinations of $\eta(t)$ and $\alpha(t)$:

1. $\eta(t) = 1, \alpha(t) = 0$: $\tilde{x}(t) = x(t)$ (no cyber-attacks);
2. $\eta(t) = 0, \alpha(t) = 0$: $\tilde{x}(t) = f(x(t))$ (only deception attacks);
3. $\alpha(t) = 1$: $\tilde{x}(t) = 0$ (active DoS attacks, blocking communication regardless of deception attacks).

For the stability analysis, the effective communication rate within the n -th attack cycle is defined as $v_n = \frac{\mu_n}{H_{n+1} - H_n}$. Furthermore, considering the energy constraints of attackers, DoS attacks cannot remain active indefinitely. Let $\mathcal{H}_M$ and $\mathcal{H}_m$ denote the maximum duration of the DoS active periods and the minimum duration of the sleeping periods, respectively, which satisfy:

$$\begin{cases} \mathcal{H}_M \geq \sup_{n \in \mathbb{N}} \{H_{n+1} - H_n - \mu_n\} \\ \mathcal{H}_m \leq \inf_{n \in \mathbb{N}} \{\mu_n\} \end{cases} \quad (11)$$

Moreover, the following assumption is introduced:

Assumption 2. (Xing et al. (2023)): For any $t_1 > t_0 \geq 0$, there exist $v_1 \geq 0$ and $\ell \geq 0$ such that the following condition holds :

$$\mathcal{N}(t_0, t_1) \leq v_1 + \frac{t_1 - t_0}{\ell} \quad (12)$$

where $\mathcal{N}(t_0, t_1)$ is the number of DoS attacks sleep/active transitions over $[t_0, t_1)$.

2.3. Memory-dependent ETM

By introducing a MDETm for each fuzzy subsystem through the following fuzzy rules:

MDETm rule s : If $\theta_1(t)$ is $\aleph_s^1$, $\theta_2(t)$ is $\aleph_s^2$, and $\dots$ and $\theta_s(t)$ is $\aleph_s^z$, then

$$\sum_{j=1}^m \mu_j e_j^s(t) \Psi_s e_j^s(t) < \lambda_s(\Lambda_k^s) \bar{x}^T(\Lambda_k^s) \Psi_s \bar{x}(\Lambda_k^s) \quad (13)$$

where m is the number of recently released packets, and $\mu_j \in [0, 1]$ are weighting parameters with $\sum_{j=1}^m \mu_j = 1$. For the s -th rule, $e_j^s(t) = x(\Lambda_k^s + \ell h) - x(\Lambda_{k-j+1}^s)$ denotes the state variation between the current sampled and the $(k - j + 1)$ -th triggered data. Additionally, Ψ_s is a positive-definite fuzzy-dependent weight matrix, and $\bar{x}(\Lambda_k^s) = \frac{1}{m} \sum_{j=1}^m x(\Lambda_{k-j+1}^s)$ denotes the average state of the m recently released packets.

Remark 2. Unlike traditional memoryless schemes, the proposed MDETm utilizes past released signals to evaluate the triggering condition. At the crest or trough of the response curve, the difference between adjacent samplings becomes very small, often causing traditional mechanisms to incorrectly drop packets. By incorporating the historical information $\bar{x}(\Lambda_k^s)$, the proposed mechanism significantly improves the releasing possibilities at these points. As a result, more packets are transmitted during severe fluctuations to expedite convergence, thereby enhancing the flexibility and adaptability of the control scheme.

Motivated by the approach in Lin et al. (2025), an adaptive threshold function $\lambda_s(\Lambda_k^s)$ is introduced:

$$\lambda_s(\Lambda_k^s) = \lambda_m + (\lambda_M - \lambda_m) \frac{2}{\pi} \text{arccot}(\mathcal{K}_s) \quad (14)$$

with

$$\mathcal{K}_s = \frac{\wp \cdot \|e_j^s(t) \cdot g_s(\theta(t))\|^2}{v_{n-1} \cdot e^{\Delta_n}}, \quad (15)$$

where $0 < \lambda_m \leq \lambda_s(\Lambda_k^s) \leq \lambda_M < 1$, $\wp$ is an adjustment parameter satisfying $\wp \geq 0$, and

$$\left\{ \begin{array}{l} \Delta_n = \sum_{v=1}^{\bar{v}} \mu_v \aleph(n - v + 1), \quad \bar{v} \leq n, \\ \aleph(n) = \begin{cases} 0, & n = 1, 2; \\ \frac{v_{n-1}}{v_{n-2}} - 1, & n \geq 3, \end{cases} \\ \sum_{v=1}^{\bar{v}} \mu_v = 1, \quad 1 \geq \mu_v \geq 0 \quad (v = 1, 2, \dots, \bar{v}). \end{array} \right.$$

The triggered time sequence of the s -th fuzzy subsystem is composed of the previous triggered time and multiple untriggered sampling intervals h , which is shown as follows:

$$\Lambda_{k+1}^s = \Lambda_k^s + \min_{l \in \mathbb{N}^*} \left\{ \ell h \left| \sum_{j=1}^m \mu_j e_j^s(t) \Psi_s e_j^s(t) \right. \right. \\ \left. \left. > \lambda_s(\Lambda_k^s) \bar{x}^T(\Lambda_k^s) \Psi_s \bar{x}(\Lambda_k^s) \right\} \quad (16)$$

All fuzzy subsystems share the same sampled data. Given that the triggering condition (16) is evaluated solely at each sampling instant, the inter-event time is strictly lower-bounded by the sampling period h . That is, $\Lambda_{k+1}^s - \Lambda_k^s \geq h > 0$, which inherently excludes Zeno behavior.

Under DoS attacks, the data communication is interrupted during the active periods $\mathcal{W}_{2,n}$, which renders the standard MDETm (16) invalid. To ensure the timely update of the system state at the end of the attacks, a forced transmission is supplemented at the exact beginning of each sleeping period H_n . The modified event-triggered sequence is formulated as:

$$\Lambda_{k,n}^s = \{\Lambda_k^s \text{ satisfying (16)} \mid \Lambda_k^s \in \mathcal{W}_{1,n}\} \cup \{H_n\} \quad (17)$$

Let $\mathcal{I}_{k,n} \triangleq [\Lambda_{k,n}^s, \Lambda_{k+1,n}^s)$ denote the holding interval between two adjacent successful triggering instants, which is divided into the following microscopic sub-intervals:

$$\mathcal{I}_{k,n} = \bigcup_{\ell=1}^{\rho_{k,n}} \mathcal{J}_{k,n}^{\ell} \cup \mathcal{J}_{k,n}^{\rho_{k,n}+1} \quad (18)$$

where $\rho_{k,n} \triangleq \sup\{\ell \in \mathbb{N} \mid \Lambda_{k,n}^s + \ell h < \Lambda_{k+1,n}^s\}$, $\mathcal{J}_{k,n}^{\ell} = [\Lambda_{k,n}^s + (\ell - 1)h, \Lambda_{k,n}^s + \ell h)$, and the residual sub-interval is $\mathcal{J}_{k,n}^{\rho_{k,n}+1} = [\Lambda_{k,n}^s + \rho_{k,n}h, \Lambda_{k+1,n}^s)$.

For $t \in \mathcal{J}_{k,n}^{\ell}$ ($\ell \in \{1, 2, \dots, \rho_{k,n} + 1\}$), the time-varying delay $\tau(t)$ and triggering error $e_j^s(t)$ are defined as:

$$\tau(t) = t - \Lambda_{k,n}^s - (\ell - 1)h \quad (19)$$

$$e_j^s(t) = x(\Lambda_{k,n}^s + (\ell - 1)h) - x(\Lambda_{k-j+1,n}^s) \quad (20)$$

Clearly, the time-varying delay satisfies $0 \leq \tau(t) < h$. Consequently, the transmitted signals can be reformulated as:

$$x(\Lambda_{k-j+1,n}^s) = x(t - \tau(t)) - e_j^s(t) \quad (21)$$

Remark 3. It should be mentioned that the memory content of the proposed MDETm includes the historical state information $\bar{x}(\Lambda_k^s)$ and historical attack intensity $\aleph(n)$, which can further reduce resource consumption and mitigate the adverse effects of DoS attacks. Specifically, the value of $\aleph(n)$ reflects the variation in the attack severity. An intensifying attack trend leads to $\aleph(n) < 0$, driving the MDETm to employ a smaller threshold for enhanced status updates, whereas a decreasing attack trend yields $\aleph(n) > 0$. Furthermore, $\aleph(n) = 0$ denotes a periodic DoS attacks occur, the threshold can still be adapted according to the preceding attack level of v_{n-1} . Additionally, The memory weighting parameters μ_i are introduced to quantify the relative importance of each historical packet. Based on the principle that newly released packets generally carry more critical information than historical ones, the weighting parameters are designed to satisfy $\mu_i \geq \mu_{i+1}$ for $i = 1, 2, \dots, m - 1$.

Remark 4. To further effectively reduce resource consumption, the proposed method further reduces resource

consumption by effectively leveraging the characteristics of the T-S fuzzy model. Different from most existing adaptive ETM (AETM), the MDETM is also designed with distinct thresholds and weighting matrices for different fuzzy rules based on the FMF. Specifically, a smaller FMF implies that the corresponding subsystem plays a less significant role in approximating the original nonlinear system; thus, a higher adaptive threshold is assigned to reduce its communication frequency. Furthermore, the proposed MDETM is more general and covers most existing ETM as special cases, which provides additional design flexibility. Specifically, when $\varphi = 0$, the MDETM reduces to a static ETM. When $\varphi \neq 0$, if $m = 1$ and $\Psi_s = \Psi$ for each fuzzy rule, the MDETM will degenerate to an AETM and lose the characteristic of fuzzy-dependence.

2.4. Attack switching closed-loop system

Due to the joint effects of the MDETM and cyber-attacks, the fuzzy controller with the latest lagged premise variables is formulated as:

$$u(t) = \sum_{s=1}^r \sum_{j=1}^m h_s(\tilde{\theta}(t)) K_{sj} \bar{x}(\Lambda_{k,n}^s), \quad t \in [\Lambda_{k,n}^s, \Lambda_{k+1,n}^s) \quad (22)$$

where $\tilde{\theta}_1(t)$ represents the system state received by the controller, and K_{sj} represents the controller gain. Moreover, it is assumed that the deviations in the membership functions satisfy the following inequalities: $g_s(\tilde{\theta}(t)) \geq \varsigma_s g_s(\theta(t))$.

The closed-loop fuzzy system can be further derived:

$$\begin{cases} \dot{x}(t) = \sum_{i=1}^r \sum_{s=1}^r h_i(\theta(t)) h_s(\tilde{\theta}(t)) \left\{ A_i x(t) + B_i \sum_{j=1}^m K_{sj} \right. \\ \quad \left. [\eta(t)(x(t - \tau(t)) - e_j^s(t)) + (1 - \eta(t))f(x(\Lambda_{k-j+1,n}^s))] \right. \\ \quad \left. + D_i w(t) \right\}, t \in \mathcal{I}_{k,n} \cap \mathcal{W}_{1,n} \\ \dot{x}(t) = \sum_{i=1}^r \sum_{s=1}^r h_i(\theta(t)) h_s(\tilde{\theta}(t)) [A_i x(t) + D_i w(t)], t \in \mathcal{W}_{2,n} \end{cases} \quad (23)$$

Additionally, the initial condition of the system is given by $x(t) = \mathcal{M}(t)$ for $t \in [-h, 0]$.

Furthermore, Fig. 2 illustrates the USV closed-loop control framework, wherein the MDETM and fuzzy control strategies are integrated against hybrid network attacks.

To facilitate subsequent calculations and derivations, three lemmas are presented as follows:

Lemma 1. (Seuret and Gouaisbaud (2013)): Defined $x(\cdot) \in [a, b] \rightarrow \mathbb{R}^n$ be a differentiable function, and $\mathbb{Z} > 0$ be any matrix, it holds that

$$(b - a) \int_a^b \dot{x}^T(s) \mathbb{Z} \dot{x}(s) ds \geq \mu_1^T \mathbb{Z} \mu_1 + 3\mu_2^T \mathbb{Z} \mu_2 \quad (24)$$

where $\mu_1 = x(b) - x(a)$ and $\mu_2 = x(b) + x(a) - \frac{2}{b-a} \int_a^b x(s) ds$.

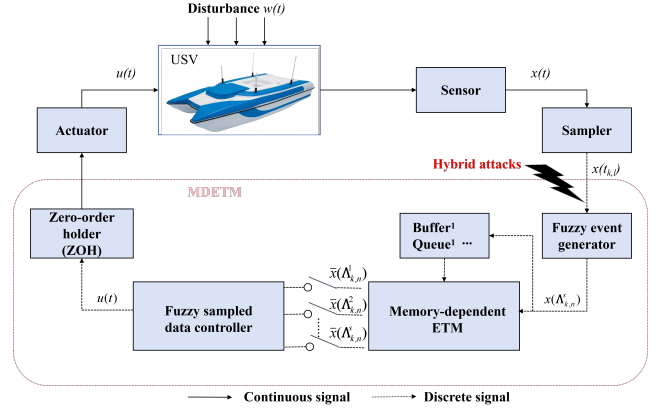


Figure 2: Control framework of USV under hybrid attacks

Lemma 2. (Zhang et al. (2018)): For a constant diagonal matrix $\Upsilon = \text{diag}\{\Upsilon_1, \Upsilon_2, \dots, \Upsilon_n\} > 0$, continuous time-varying delays satisfying $0 < \tau_1 \leq \tau_i(t) \leq h$, and continuously differentiable functions $x_i(t)$ for $i = 1, 2, \dots, n$, if there exist scalars $r_1, r_2, \dots, r_n$ satisfying the following conditions:

$$\begin{bmatrix} \Upsilon_i & r_i \\ * & \Upsilon_i \end{bmatrix} \geq 0, \quad i = 1, 2, \dots, n \quad (25)$$

then the following integral inequality holds:

$$\begin{aligned} & - \sum_{i=1}^n (h - \tau_1) \Upsilon_i \int_{t-h}^{t-\tau_1} \dot{x}_i^2(s) ds \\ & \leq - \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix}^T \begin{bmatrix} \Upsilon & R \\ * & \Upsilon \end{bmatrix} \begin{bmatrix} \mathbf{v}_1(t) \\ \mathbf{v}_2(t) \end{bmatrix} \end{aligned} \quad (26)$$

where $\mathbf{v}_j(t) = [\mathbf{v}_{j1}(t), \mathbf{v}_{j2}(t), \dots, \mathbf{v}_{jn}(t)]^T$ for $j = 1, 2$, with their elements given by $\mathbf{v}_{1i}(t) = x_i(t - \tau_1) - x_i(t - \tau_i(t))$ and $\mathbf{v}_{2i}(t) = x_i(t - \tau_i(t)) - x_i(t - h)$ respectively, and $R = \text{diag}\{r_1, r_2, \dots, r_n\}$.

Lemma 3 (Zhang et al. (2017)). Given a scalar $d \in (0, 1)$, vectors $\mathcal{M}_1, \mathcal{M}_2 \in \mathbb{R}^n$, positive definite matrices $\mathcal{G}_1, \mathcal{G}_2 \in \mathbb{R}^{n \times n}$, and arbitrary matrices $Y_1, Y_2 \in \mathbb{R}^{n \times n}$, the following inequality holds:

$$\begin{aligned} & \frac{1}{d} \mathcal{M}_1^T \mathcal{G}_1 \mathcal{M}_1 + \frac{1}{1-d} \mathcal{M}_2^T \mathcal{G}_2 \mathcal{M}_2 \\ & \geq \mathcal{M}_1^T [\mathcal{G}_1 + (1-d)(\mathcal{G}_1 - Y_1 \mathcal{G}_2^{-1} Y_1^T)] \mathcal{M}_1 + 2\mathcal{M}_1^T [dY_1 \\ & \quad + (1-d)Y_2] \mathcal{M}_2 + \mathcal{M}_2^T [\mathcal{G}_2 + d(\mathcal{G}_2 - Y_2^T \mathcal{G}_1^{-1} Y_2)] \mathcal{M}_2. \end{aligned}$$

Definition 1. Consider the closed-loop system (23). If the system is asymptotically stable when $\omega(t) = 0$, and under zero initial conditions, the following inequality holds:

$$\int_0^\infty y^T(\tau) y(\tau) d\tau \leq \gamma^2 \int_0^\infty \omega^T(\tau) \omega(\tau) d\tau,$$

then the system is said to be asymptotically stable with an H_∞ performance index $\gamma > 0$.

Definition 2. (Liu et al. (2021)): For known scalars $\chi_1 > 0$, $\chi_2 > 0$, and β_0 , if $E\{\|x(t)\|^2\} \leq \chi_1 e^{-\chi_2 t} E\{\|\beta_0\|^2\}$ holds, the system is regarded to be EMSS with the initial condition.

3. Main Results

In this section, exponentially stability analysis of closed-loop fuzzy system (23) under hybrid cyber-attacks is established. For the sake of simplicity, the following notations are introduced:

$$r_\kappa^1 = [0_{n \times (\kappa-1)n} \quad I_{n \times n} \quad 0_{n \times (8+2m-\kappa)n}] , \kappa \in \{1, 2, 3, 6, \dots, 10\}$$

$$r_0^1 = 0_{n \times 10n}, r_4^1 = [0_{n \times 3n} \quad I_{n \times mn} \quad 0_{n \times (5+m)n}]$$

$$r_5^1 = [0_{n \times (3+m)n} \quad I_{n \times mn} \quad 0_{n \times (4+m)n}] ,$$

$$r_d^1 = [0_{n \times (d-1)n} \quad I_{n \times n} \quad 0_{n \times (8-d)n}] , d \in \{1, 2, \dots, 8\}$$

$$r_0^2 = 0_{n \times 8n}, \quad \mathfrak{T}_1 = t - t_k, \mathfrak{T}_2 = t_{k+1} - t$$

$$v_1(t) = \frac{1}{\mathfrak{T}_1} \int_{t_k}^t x(s) ds, v_2(t) = \frac{1}{\mathfrak{T}_2} \int_t^{t_{k+1}} x(s) ds$$

$$\zeta(t) = \text{col}\{x(t), x_\tau(t), x_h(t), E(t), F(t), w(t)\}$$

$$\xi_1(t) = \text{col}\{\zeta(t), x(t_k), x(t_{k+1}), v_1(t), v_2(t)\}$$

$$\xi_2(t) = \text{col}\{x(t), x_\tau(t), x_h(t), w(t), x(t_k), x(t_{k+1}), v_1(t), v_2(t)\}$$

$$x_\tau(t) = x(t - \tau(t)), \quad x_h(t) = x(t - h)$$

$$E(t) = [e_1^s(t), \dots, e_m^s(t)]$$

$$F(t) = [f(x(\Lambda_{k,n}^s)), \dots, f(x(\Lambda_{k-m+1,n}^s))]$$

3.1. Stability analysis

Theorem 1. For a given sampling period h , the DoS parameters $\mathcal{H}_M$ and $\mathcal{H}_m$, the positive scalars δ_1, δ_2, a_1 and a_2 satisfying $a_1 a_2 > 1$, the deception attack probability $\bar{\eta}$, the controller gains K_{sj} , and the constant matrix Φ , the system (23) is EMSS if there exist positive-definite matrices $P_i^\zeta, Q_{\ell i}^\zeta, O_{\mathcal{L}}^\zeta$, MDET weight matrices Ψ_s , arbitrary matrices $X_\ell^\zeta, Y_\ell^\zeta$, and symmetric matrices N_q ($q = 1, 2, 3$) with appropriate dimensions, such that the following inequalities hold for any $\dot{g}_k(t) \in [-\mathcal{D}_k, \mathcal{D}_k]$, where $l \neq k, k, l, i, s \in \{1, 2, \dots, r\}, j \in \{1, 2, \dots, m\}$, and $\ell, i, \zeta, \mathcal{L} \in \{1, 2\}$:

$$\begin{cases} P_l^\zeta + N_1 - P_k^\zeta \geq 0 \\ Q_{1l}^\zeta + N_2 - Q_{1k}^\zeta \geq 0 \\ Q_{2l}^\zeta + N_3 - Q_{2k}^\zeta \geq 0 \end{cases} \quad (27)$$

$$\dot{g}_k(t)N_2 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(Q_{1l}^\zeta + N_2 - Q_{1k}^\zeta) - X_1^\zeta \leq 0 \quad (28)$$

$$\dot{g}_k(t)N_3 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(Q_{2l}^\zeta + N_3 - Q_{2k}^\zeta) - X_2^\zeta \leq 0 \quad (29)$$

$$\begin{cases} P_i^1 - a_2 P_i^2 \leq 0, & e^{-2(\delta_1 + \delta_2)h} P_i^2 - a_1 P_i^1 \leq 0 \\ Q_{\ell i}^1 - a_2 Q_{\ell i}^2 \leq 0, & X_1^1 - a_2 X_1^2 \leq 0 \\ Q_{\ell i}^2 - a_1 Q_{\ell i}^1 \leq 0, & X_2^1 - a_2 X_2^2 \leq 0 \end{cases} \quad (30)$$

$$\begin{bmatrix} Q_{2i}^1 & Z_1 \\ * & Q_{2i}^1 \end{bmatrix} > 0, \quad \begin{bmatrix} Q_{2i}^2 & Z_2 \\ * & Q_{2i}^2 \end{bmatrix} > 0 \quad (31)$$

$$\Xi_{is}^1 = \begin{bmatrix} \Lambda_{11}^1 + h\Gamma_i^1 & \Lambda_{12}^1 \\ * & \Lambda_{22}^1 \end{bmatrix} < 0 \quad (32)$$

$$\Xi_{is}^2 = \begin{bmatrix} \Lambda_{11}^2 + h\Gamma_i^2 & \Lambda_{12}^2 \\ * & \Lambda_{22}^2 \end{bmatrix} < 0 \quad (33)$$

where

$$\Lambda_{11}^1 = -e^{-2\delta_1 h} \Pi_1^T O_1^1 \Pi_1^1 - e^{-2\delta_1 h} \Pi_2^T O_2^1 \Pi_2^1 + W_2^T \Lambda_3 W_2$$

$$\Lambda_{11}^2 = -e^{2\delta_2 h} \Pi_2^T O_2^2 \Pi_2^2 - e^{2\delta_2 h} \Pi_2^T O_2^2 \Pi_2^2 + W_3^T \Lambda_4 W_3$$

$$\Gamma_l^1 = \text{sym} \left\{ -\frac{1}{h} e^{-2\delta_1 h} \Pi_1^T Y_2 \Pi_2^1 \right\} - \frac{1}{h} e^{-2\delta_1 h} \Pi_l^T O_l^1 \Pi_l^1 \\ + h W_2^T (G_1^T O_l^1 G_1 + G_2^T O_l^1 G_2) W_2$$

$$\Gamma_l^2 = \text{sym} \left\{ -\frac{1}{h} e^{2\delta_2 h} \Pi_2^T Y_2 \Pi_2^2 \right\} - \frac{1}{h} e^{2\delta_2 h} \Pi_l^T O_l^2 \Pi_l^2 \\ + h W_2^T (G_1^T O_l^2 G_1 + G_2^T O_l^2 G_2) W_2$$

$$\Lambda_1 = \dot{g}_k(t)N_1 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(P_l^1 + N_1 - P_k^1)$$

$$\Lambda_2 = \dot{g}_k(t)N_1 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(P_l^2 + N_1 - P_k^2)$$

$$\Lambda_3 = \begin{bmatrix} \Lambda_{111}^1 & \Lambda_{112}^1 \\ * & \Lambda_{113}^1 \end{bmatrix}, \Lambda_{111}^1 = \begin{bmatrix} \pi_{11}^1 & \pi_{12}^1 & \pi_{13}^1 \\ * & \pi_{22}^1 & \pi_{23}^1 \\ * & * & \pi_{33}^1 \end{bmatrix}$$

$$\Lambda_{112}^1 = [\Sigma_{12} \quad \Sigma_{13} \quad \Sigma_{14}], \Lambda_{113}^1 = \text{diag}\{\Sigma_{22}, \Sigma_{33}, \Sigma_{44}\}$$

$$N_{1s} = \sum_{j=1}^m B K_{sj}, N_{2s} = B[K_{s1}, \dots, K_{sm}]$$

$$\pi_{11}^1 = 2\delta_1 P_i^1 + \text{sym}(P_i^1 A_i) + Q_{1i}^1 - e^{-2\delta_1 h} Q_{2i}^1 + h X_1^1 + \Lambda_1$$

$$\pi_{12}^1 = \bar{\eta} P_i^1 N_{1s} + e^{-2\delta_1 h} (Q_{2i}^1 + Z_1), \pi_{13}^1 = -e^{-2\delta_1 h} Z_1$$

$$\pi_{22}^1 = e^{-2\delta_1 h} (-2Q_{2i}^1 - \text{sym}(Z_1)) + \lambda_M \Psi_s$$

$$\pi_{23}^1 = e^{-2\delta_1 h} (Q_{2i}^1 + Z_1), \pi_{33}^1 = -e^{-2\delta_1 h} (Q_{1i}^1 + Q_{2i}^1)$$

$$\Sigma_{12} = \begin{bmatrix} -\bar{\eta} P_i^1 N_{2s} \\ -\frac{\lambda_M}{m} \Psi_s \\ 0 \end{bmatrix}, \Sigma_{13} = \begin{bmatrix} (1-\bar{\eta}) P_i^1 N_{2s} \\ 0 \end{bmatrix}, \Sigma_{14} = \begin{bmatrix} P_i^1 D_i \\ 0 \end{bmatrix}$$

$$\Sigma_{22} = \text{diag}\{(\frac{\lambda_M}{m^2} - \mu_1) \Psi_s, \dots, (\frac{\lambda_M}{m^2} - \mu_m) \Psi_s\}$$

$$\Sigma_{33} = \text{diag}\{-I, \dots, -I\}, \Sigma_{44} = -\gamma^2 I$$

$$\Lambda_{12}^1 = [\Lambda_{121}^1 \quad \Lambda_{122}^1], \Lambda_{22}^1 = [\Lambda_{221}^1 \quad \Lambda_{222}^1]$$

$$\Lambda_{121}^1 = [C_i^T \quad \Phi^T \quad h G_1^T Q_{2i}^1 \quad h G_2^T Q_{2i}^1]$$

$$\Lambda_{122}^1 = \begin{bmatrix} \frac{\sqrt{2}h}{2} h G_1^T X_2^1 & \frac{\sqrt{2}h}{2} h G_2^T X_2^1 & \Pi_1^T Y_1 \end{bmatrix}$$

$$\Lambda_{221}^1 = \text{diag}\{-I, -\bar{\eta} I, -Q_{2i}^1, -Q_{2i}^1\}$$

$$\Lambda_{222}^1 = \text{diag}\{-X_2^1, -X_2^1, -e^{-2\delta_1 h} O_2^{\zeta T}\}$$

$$\Lambda_4 = \begin{bmatrix} \pi_{11}^2 & Q_{2i}^2 + Z_2 & -Z_2 & P_i^2 D_i \\ * & -2Q_{2i}^2 - \text{sym}(Z_2) & Q_{2i}^2 + Z_2 & 0 \\ * & * & -e^{2\delta_2 h} Q_{1i}^2 - Q_{2i}^2 & 0 \\ * & * & * & -\gamma^2 I \end{bmatrix}$$

$$\pi_{11}^2 = -2\delta_2 P_i^2 + \text{sym}(P_i^2 A_i) + Q_{1i}^2 - Q_{2i}^2 + h X_1^2 + \Lambda_2$$

$$\Lambda_{12}^2 = [\Lambda_{121}^2 \quad \Lambda_{122}^2 \quad \Lambda_{123}^2 \quad \Lambda_{124}^2]$$

$$\Lambda_{22}^2 = \text{diag}\{-I, -Q_{2i}^2, -X_2^2, O_2^{\zeta T}\}$$

$$\begin{aligned}\Lambda_{121}^2 &= [C_i \ 0 \ 0 \ 0]^T, \Lambda_{124}^2 = [\Pi_2^T Y_2]^T, \\ \Lambda_{122}^2 &= [hQ_{2i}^2 A_i \ 0 \ 0 \ hQ_{2i}^2 D_i]^T \\ \Lambda_{123}^2 &= \left[\frac{\sqrt{2h}}{2} hX_2^2 A_i \ 0 \ 0 \ \frac{\sqrt{2h}}{2} hX_2^2 D_i \right]^T \\ \Pi_1^1 &= \text{col}\{r_1^1 - r_7^1, r_1^1 + r_7^1 - 2r_9^1\} \\ \Pi_2^1 &= \text{col}\{r_8^1 - r_1^1, r_1^1 + r_8^1 - 2r_{10}^1\} \\ \Pi_1^2 &= \text{col}\{r_1^2 - r_5^2, r_1^2 + r_5^2 - 2r_7^2\} \\ \Pi_2^2 &= \text{col}\{r_6^2 - r_1^2, r_1^2 + r_6^2 - 2r_8^2\}, W_1 = \text{col}\{r_1^1, r_2^1, r_3^1\} \\ W_2 &= \text{col}\{r_1^1, r_2^1, r_3^1, r_4^1, r_5^1, r_6^1\}, W_3 = \text{col}\{r_1^2, r_2^2, r_3^2, r_4^2\}\end{aligned}$$

Proof: For system (6) under MDET (16), the following LKF is selected:

$$V_\zeta(t) = \sum_{i=1}^4 V_{\zeta i}(t), \zeta \in \{1, 2\} \quad (34)$$

where

$$\begin{aligned}V_{\zeta 1}(t) &= \sum_{i=1}^r g_i(\theta(t)) x^T(t) P_i^\zeta x(t) \\ V_{\zeta 2}(t) &= \sum_{i=1}^r g_i(\theta(t)) \int_{t-h}^t x^T(s) \rho(\cdot) Q_{1i}^\zeta x(s) ds \\ &\quad + \sum_{i=1}^r g_i(\theta(t)) h \int_{-h}^0 \int_{t+\theta}^t \dot{x}^T(s) \rho(\cdot) Q_{2i}^\zeta \dot{x}(s) ds d\theta \\ V_{\zeta 3}(t) &= \mathfrak{Z}_2 h \int_{t_k}^t \dot{x}^T(s) \rho(\cdot) O_1^\zeta \dot{x}(s) ds \\ &\quad - \mathfrak{Z}_1 h \int_t^{t_{k+1}} \dot{x}^T(s) \rho(\cdot) O_2^\zeta \dot{x}(s) ds \\ V_{\zeta 4}(t) &= \int_{t-h}^t \int_\theta^t x^T(s) \rho(\cdot) X_1^\zeta x(s) ds d\theta \\ &\quad + h \int_{-h}^0 \int_\theta^0 \int_{t+u}^t \dot{x}^T(s) \rho(\cdot) X_2^\zeta \dot{x}(s) ds du d\theta,\end{aligned}$$

with $\rho(\cdot) = e^{(-1)^\zeta 2\delta_\zeta(t-s)}$, $\rho_1 = e^{-2\delta_1 h}$, $\rho_2 = e^{2\delta_2 h}$.

Let $P_g^\zeta = \sum_{i=1}^r g_i(t) P_i^\zeta$, $Q_{\ell g}^\zeta = \sum_{i=1}^r g_i(t) Q_{\ell i}^\zeta$, with $\ell \in \{1, 2\}$. In the subsequent analysis, the two distinct scenarios of $\zeta = 1$ and $\zeta = 2$ are investigated separately.

Case 1: When $\zeta = 1$, calculating the mathematical expectation of the time derivative of $V_1(t)$ for $t \in \mathcal{W}_{1,n}$, one obtains:

$$\begin{aligned}E\{\dot{V}_{11}(t)\} &= 2E\left\{x^T(t) P_g^1 \dot{x}(t)\right\} + x^T(t) \dot{P}_g^1 x(t) \\ E\{\dot{V}_{12}(t)\} &= -2\delta_1 E\{V_{12}(t)\} + E\left\{h^2 \dot{x}^T(t) Q_{2g}^1 \dot{x}(t)\right\} \\ &\quad + x^T(t) Q_{1g}^1 x(t) - e^{-2\delta_1 \tau_2} x_{\tau_2}^T(t) Q_{1g}^1 x_{\tau_2}(t) \\ &\quad - h e^{-2\delta_1 h} \int_{t-h}^t \dot{x}^T(s) Q_{2g}^1 \dot{x}(s) ds \\ &\quad + \int_{t-h}^t x^T(s) \rho_1 \dot{Q}_{1g}^1 x(s) ds\end{aligned}$$

$$\begin{aligned}&+ h \int_{-h}^0 \int_{t+\theta}^t \dot{x}^T(s) \rho_1 \dot{Q}_{2g}^1 \dot{x}(s) ds d\theta \\ E\{\dot{V}_{13}(t)\} &\leq -2\delta_1 E\{V_{13}(t)\} + \mathfrak{Z}_2 h E\left\{\dot{x}^T(t) O_1^1 \dot{x}(t)\right\} \\ &\quad + \mathfrak{Z}_1 h E\left\{\dot{x}^T(t) O_2^1 \dot{x}(t)\right\} \\ &\quad - e^{-2\delta_1 h} h \int_{t_k}^t \dot{x}^T(s) O_1^1 \dot{x}(s) ds \\ &\quad - e^{-2\delta_1 h} h \int_t^{t_{k+1}} \dot{x}^T(s) O_2^1 \dot{x}(s) ds \\ E\{\dot{V}_{14}(t)\} &= -2\delta_1 E\{V_{14}(t)\} + \frac{h^3}{2} E\left\{\dot{x}^T(t) X_2^1 \dot{x}(t)\right\} \\ &\quad - \int_{t-h}^t x^T(s) \rho(\cdot) X_1^1 x(s) ds + h x^T(t) X_1^1 x(t) \\ &\quad - h \int_{-h}^0 \int_{t+\theta}^t \dot{x}^T(s) \rho(\cdot) X_2^1 \dot{x}(s) ds d\theta\end{aligned}$$

Furthermore, substituting (23) into the mathematical expectations of $\dot{V}_{11}(t)$ and $\dot{V}_{12}(t)$ yields the following results:

$$2E\left\{x^T(t) P_g^1 \dot{x}(t)\right\} = 2x^T(t) P_g^1 G_1 \zeta(t) \quad (35)$$

$$\begin{aligned}&h^2 E\left\{\dot{x}^T(t) Q_{2g}^1 \dot{x}(t)\right\} \\ &= h^2 \zeta^T(t) \left(G_1^T Q_{2g}^1 G_1 + G_2^T Q_{2g}^1 G_2\right) \zeta(t)\end{aligned} \quad (36)$$

Moreover, the corresponding terms in $\dot{V}_{13}(t)$ and $\dot{V}_{14}(t)$ can be handled similarly to (36), where

$$\begin{aligned}G_1 &= [A_i \ \bar{\eta} N_{1j} \ 0 \ -\bar{\eta} N_{2j} \ (1 - \bar{\eta}) N_{2j} \ D_i], \\ G_2 &= [0 \ \rho N_{1j} \ 0 \ -\rho N_{2j} \ -\rho N_{2j} \ 0].\end{aligned}$$

By applying Jensen's inequality and Lemma (2), the following deduction can be obtained:

$$\begin{aligned}&- h e^{-2\delta_1 h} \int_{t-h}^t \dot{x}^T(s) Q_{2g}^1 \dot{x}(s) ds \\ &\leq \zeta_1^T(t) \left[e^{-2\delta_1 h} (-W_1^T \hat{Q}_{2i}^1 W_1)\right] \zeta_1(t)\end{aligned} \quad (37)$$

From Lemmas (1) - (3), the integral terms contained in $E\{\dot{V}_{13}(t)\}$ are handled:

$$\begin{aligned}&- e^{-2\delta_1 h} h \int_{t_k}^t \dot{x}^T(s) O_1^1 \dot{x}(s) ds \\ &- e^{-2\delta_1 h} h \int_t^{t_{k+1}} \dot{x}^T(s) O_2^1 \dot{x}(s) ds \\ &\leq -e^{-2\delta_1 h} \zeta(t)^T \left[\Pi_1^T \left(O_1^{1'} + \frac{\mathfrak{Z}_2}{h} (O_1^{1'} - Y_1 O_2^{1'-1} Y_1^T)\right) \Pi_1^1\right. \\ &\quad + \Pi_2^T \left(O_2^{1'} + \frac{\mathfrak{Z}_1}{h} (O_2^{1'} - Y_2^T O_1^{1'-1} Y_2)\right) \Pi_2^1 \\ &\quad \left. + \text{sym}\left\{\Pi_1^T \left(\frac{\mathfrak{Z}_1 Y_1}{h} + \frac{\mathfrak{Z}_2 Y_2}{h}\right) \Pi_2^1\right\}\right] \zeta(t)\end{aligned} \quad (38)$$

where

$$O_1^{1'} = \text{diag}\{O_1^1, 3O_1^1\}, O_2^{1'} = \text{diag}\{O_2^1, 3O_2^1\}$$

$$\hat{Q}_{2i}^1 = \begin{bmatrix} -Q_{2i}^1 & Q_{2i}^1 + Z_1 & -Z_1^T \\ * & -2Q_{2i}^1 - \text{sym}(Z_1) & Q_{2i}^1 + Z_1^T \\ * & * & -Q_{2i}^1 \end{bmatrix}.$$

According to Assumption 1, it yields that

$$\bar{\eta} x^T(t) \Phi^T \Phi x(t) - \bar{\eta} f^T(x(t)) f(x(t)) \geq 0. \quad (39)$$

Recalling MDETM (13) and (16), the following condition will be obtained:

$$\begin{aligned} & \sum_{s=1}^r \sum_{j=1}^m \mu_i e_j^s(t)^T \Psi_s e_j^s(t) \\ & \leq \sum_{s=1}^r \sum_{j=1}^m \frac{1}{m^2} \lambda_M(x_\tau(t) - e_j^s(t))^T \Psi_s (x_\tau(t) - e_j^s(t))^T \end{aligned} \quad (40)$$

Drawing inspiration from Li et al. (2025), the FMF time derivatives $\dot{g}_i(t)$ within the quadratic and integral terms are analyzed. A zero equality $x^T(t) \left(\sum_{i=1}^r \dot{g}_i(t) N_1 \right) x(t) = 0$ sharing the same structure as $x^T(t) \dot{P}_{g(t)} x(t)$ is introduced, where N_1 is a symmetric free-weighting matrix. Thus, the following equality can be expanded:

$$x^T(t) \dot{P}_{g(t)}^1 x(t) = x^T(t) \left(\sum_{i=1}^r \dot{g}_i(t) (P_i^1 + N_1) \right) x(t). \quad (41)$$

From (7), it follows that $\dot{g}_k(t) = -\sum_{i=1, i \neq k}^r \dot{g}_i(t)$. Considering the constraint $|\dot{g}_i(t)| \leq \mathcal{D}_l$ and applying (27), the following inequality is obtained:

$$\begin{aligned} & x^T(t) \left(\sum_{i=1}^r \dot{g}_i(t) (P_i^1 + N_1) \right) x(t) \\ & = x^T(t) \left(\dot{g}_k(t) N_1 + \dot{g}_k(t) P_k^1 + \sum_{i=1, i \neq k}^r \dot{g}_i(t) (P_i^1 + N_1) \right) x(t) \\ & = x^T(t) \left(\dot{g}_k(t) N_1 + \sum_{i=1, i \neq k}^r \dot{g}_i(t) (P_i^1 + N_1 - P_k^1) \right) x(t) \\ & \leq x^T(t) \left(\dot{g}_k(t) N_1 + \sum_{i=1, i \neq k}^r \mathcal{D}_l (P_i^1 + N_1 - P_k^1) \right) x(t) \end{aligned} \quad (42)$$

Similarly, by introducing free matrices N_2 and N_3 that correspond to the structure of the fuzzy-dependent terms in $E\{\dot{V}_{12}(t)\}$, and applying conditions (28) and (29), it yields:

$$\begin{aligned} & \int_{t-h}^t x^T(s) \rho_1 \left(\dot{g}_k(t) N_2 + \right. \\ & \left. \sum_{l=1, l \neq k}^r \mathcal{D}_l (Q_{1l}^1 + N_2 - Q_{1k}^1) - X_1^1 \right) x(s) ds < 0 \end{aligned} \quad (43)$$

$$\begin{aligned} & h \int_{-h}^0 \int_{t+\theta}^t \dot{x}^T(s) \rho_1 \left(\dot{g}_k(t) N_3 + \right. \\ & \left. \sum_{l=1, l \neq k}^r \mathcal{D}_l (Q_{1l}^1 + N_3 - Q_{1k}^1) - X_2^1 \right) \dot{x}(s) ds d\theta < 0 \end{aligned} \quad (44)$$

Based on the aforementioned analysis and the Schur complement lemma, one obtains:

$$\begin{aligned} & E\{\dot{V}_1(t)\} + y^T(t)y(t) - \tilde{\gamma}^2 \omega^T(t)\omega(t) \\ & \leq -2\delta_1 E\{V_1(t)\} + \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\tilde{\theta}(t)) \tilde{\xi}_1^T(t) \Xi_{is}^1 \tilde{\xi}_1(t) \end{aligned} \quad (45)$$

where

$$\begin{aligned} \Xi_{is}^1 &= \frac{\mathfrak{Z}_2}{h} (\Lambda_{11}^1 + h\mathcal{Z}_1) + \frac{\mathfrak{Z}_1}{h} (\Lambda_{11}^1 + h\mathcal{Z}_2), \\ \mathcal{Z}_1 &= \Gamma_1^1 + \frac{1}{h} e^{-2\delta_1 h} \Pi_1^1 Y_1^T O_{2i}^1 Y_1 \Pi_1^1, \\ \mathcal{Z}_2 &= \Gamma_2^1 + \frac{1}{h} e^{-2\delta_1 h} \Pi_2^1 Y_2^T O_{1i}^1 Y_2 \Pi_2^1, \\ \tilde{\xi}_1^T(t) &= [\xi_1^T(t), I, I, I, I, I, I]^T. \end{aligned}$$

It can be deduced that inequality $\Xi_{is}^1 < 0$ holds if inequalities (27)-(33) hold. Thus, over the interval $t \in \mathcal{W}_{1,n}$, it follows that $E\{\dot{V}_1(t)\} \leq -2\delta_1 E\{V_1(t)\}$.

Case 2: When $\zeta = 2$, calculating the mathematical expectation of the time derivative of $V_2(t)$ for $t \in \mathcal{W}_{2,n}$, $n \in \mathcal{N}$, the expression for $V_2(t)$ is obtained as follows:

$$\begin{aligned} & E\{\dot{V}_2(t)\} + y^T(t)y(t) - \tilde{\gamma}^2 \omega^T(t)\omega(t) \\ & \leq 2\delta_2 E\{V_2(t)\} + \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\tilde{\theta}(t)) \tilde{\xi}_2^T(t) \Xi_{is}^2 \tilde{\xi}_2(t) \end{aligned} \quad (46)$$

where

$$\begin{aligned} \Xi_{is}^2 &= \frac{\mathfrak{Z}_2}{h} (\Lambda_{11}^2 + h\mathcal{Z}_3) + \frac{\mathfrak{Z}_1}{h} (\Lambda_{11}^2 + h\mathcal{Z}_4), \\ \mathcal{Z}_3 &= \Gamma_1^2 + \frac{1}{h} e^{2\delta_2 h} \Pi_1^2 Y_1^T O_{2i}^2 Y_1 \Pi_1^2, \\ \mathcal{Z}_4 &= \Gamma_2^2 + \frac{1}{h} e^{2\delta_2 h} \Pi_2^2 Y_2^T O_{1i}^2 Y_2 \Pi_2^2, \\ \tilde{\xi}_2^T(t) &= [\xi_2^T(t), I, I, I, I]^T. \end{aligned}$$

By using $\Xi_{is}^2 < 0$, over the interval $t \in \mathcal{W}_{2,n}$, it follows that $E\{\dot{V}_2(t)\} \leq 2\delta_2 E\{V_2(t)\}$.

Based on the abovementioned analysis, for $t \in \mathcal{W}_{1,n}$, one can obtain the following:

$$E\{V_1(t)\} \leq e^{-2\delta_1(t-H_n)} E\{V_1(H_n)\} \quad (47)$$

and, $\forall t \in \mathcal{W}_{2,n}$,

$$E\{V_2(t)\} \leq e^{2\delta_2(t-H_n-\mu_n)} E\{V_2(H_n + \mu_n)\}. \quad (48)$$

from (30), one can derive that

$$E\{V_1(H_n)\} \leq a_2 E\{V_2(H_n^-)\}. \quad (49)$$

Similarly,

$$E\{V_2(H_n + \mu_n)\} \leq a_1 e^{2(\delta_1 + \delta_2)h} E\{V_1(H_n + \mu_n)^-\}. \quad (50)$$

According to (47)-(50) and mathematical iteration method, for $t \in [H_n, H_n + \mu_n]$, $N(0, t) = n$, and the following results are obtained:

$$\begin{aligned} E\{V_1(t)\} &\leq a_2 e^{-2\delta_1(t-H_n)} E\{V_2(H_n^-)\} \\ &\leq a_1 a_2 e^{-2\delta_1(t-H_n)} e^{2(\delta_1 + \delta_2)h} e^{2\delta_2(H_n - H_{n-1} - \mu_{n-1})} \\ &\quad E\{V_1(H_{n-1} - \mu_{n-1})^-\} \\ &\quad \vdots \\ &\leq (a_1 a_2)^n e^{2n(\delta_1 + \delta_2)h} e^{2n\delta_2 H_M} e^{-2n\delta_1 H_M} E\{V_1(0)\} \\ &\leq e^{-\varpi n} E\{V_1(0)\} \end{aligned} \quad (51)$$

where $\varpi = 2\delta_1 H_M - 2\delta_2 H_M - 2(\delta_1 + \delta_2)h - \ln a_1 a_2$.

Note that $H_n \leq t < H_n + \mu_n$, based on Assumption (2),

$$v_1 + \frac{t - H_n}{\ell} < N(0, t) \leq v_1 + \frac{t}{\ell}. \quad (52)$$

Then, it follows that

$$E\{V_1(t)\} < e^{-\varpi v_1 + \frac{\varpi H_M}{\ell}} \cdot e^{-\frac{\varpi}{\ell} t} E\{V_1(0)\}. \quad (53)$$

When $t \in [H_n + \mu_n, H_{n+1}]$, $v_1 + \frac{t}{\ell} < n + 1$ is derived from Definition (2), then

$$E\{V_2(t)\} \leq \frac{1}{a_2} e^{-\varpi(n+1)} \cdot e^{-\frac{\varpi}{\ell} t} E\{V_1(0)\}. \quad (54)$$

Next, it holds that

$$V(t) < b \cdot e^{-\frac{\varpi}{\ell} t} V_1(0), \quad (55)$$

where $b = \max \left\{ e^{-\varpi v_1 + \frac{\varpi H_M}{\ell}}, \frac{1}{a_2} e^{-\varpi(n+1)} \right\}$.

Remark 5. Specifically, for $i = 3$, the looped-functional terms V_{ζ_i} condition ($V(t_k) = V(t_k + 1) = 0$) can be fulfilled at each sampling instant. Meanwhile, the positive definiteness of $V(t)$ is guaranteed by the definition of $V_{\zeta_i}(t_k)$ ($i = 1, 2, 4$). Furthermore, over the interval $t \in [t_k, t_{k+1})$, based on inequalities (50)-(51), the condition $V_{\zeta}(t) > 0$ is satisfied at each sampling instant. It follows that $V(t)$ constitutes a valid LKF, and the positive definiteness constraints imposed on the looped-functional terms V_{ζ_3} can be effectively relaxed. This property enhances the flexibility of the stability condition, thereby facilitating the achievement of superior system performance.

Based on the above discussion, from the definition of $V(t)$, it can be readily derived that

$$\begin{cases} V_1(t) \geq V_{11}(0) \geq b_1 \|x(t)\|^2, \\ V_1(0) \leq V_{11}(0) + V_{12}(0) + V_{14}(0) \leq b_2 \|x_0\|^2, \end{cases} \quad (56)$$

where $b_1 = \min \{ \lambda_{\min}(P_i^1), \lambda_{\min}(P_i^2) \}$ and $b_2 = \lambda_{\max}(P_i^1) + h \lambda_{\max}(Q_{1i}^1) + \frac{h^3}{2} \lambda_{\max}(Q_{2i}^1) + \frac{h^2}{2} \lambda_{\max}(X_1^1) + \frac{h^4}{6} \lambda_{\max}(X_2^1)$. Furthermore, it follows that

$$\|x(t)\| \leq \sqrt{\frac{b \cdot b_2}{b_1}} e^{-\frac{\varpi}{2\ell} t} \|x_0\|. \quad (57)$$

Define $\chi = \sqrt{\frac{b \cdot b_2}{b_1}}$ and the decay rate $v_2 = \frac{\varpi}{2\ell}$, and the system (23) is thus deduced to be EMSS.

By (46), (49), and (50), for $\forall t \in [0, H_{n+1}]$

$$\begin{aligned} &\sum_{k=0}^n \int_{H_k}^{H_k + \mu_k} e^{-2\delta_1(H_k - t)} E(\dot{V}_1(t) + 2\delta_1 V_1(t)) dt \\ &+ \sum_{k=0}^n \int_{H_k + \mu_k}^{H_{k+1}} a_2 e^{2\delta_2(H_{k+1} - t)} E(\dot{V}_2(t) - 2\delta_2 V_2(t)) dt \\ &\geq V_1(H_{n+1}) - V_1(0) + \sum_{k=0}^n \left(\left(\frac{1}{a_1} e^{-2(\delta_1 + \delta_2)h} e^{2\delta_1 \mu_k} \right. \right. \\ &\quad \left. \left. - a_2 e^{2\delta_2(H_{k+1} - H_k - \mu_k)} \right) V_2(H_k + \mu_k) \right). \end{aligned} \quad (58)$$

Given the decay rate $v_2 = \frac{\varpi}{2\ell} > 0$ with $\ell \in \mathbb{R} > 0$, the parameter constraint is obtained as

$$\varpi = 2\delta_1 H_M - 2\delta_2 H_M - 2\delta_1 \delta_2 h - \ln a_1 a_2 > 0.$$

Moreover, due to $a_1 a_2 > 1$, then

$$\frac{1}{a_1} e^{-2(\delta_1 + \delta_2)h + 2\delta_1 H_M} > a_2 e^{2\delta_2 H_M}. \quad (59)$$

Therefore, from (45)-(46) and $V_1(0) = 0$, thus

$$\begin{aligned} &\sum_{k=0}^n \int_{H_k}^{H_{k+1}} \min\{1, a_2\}^k y^T(t) y(t) dt \\ &< \sum_{k=0}^n \int_{H_k}^{H_{k+1}} \max\{e^{2\delta_1 H_M}, e^{2\delta_2 H_M}\} \tilde{\gamma}^2 \omega^T(t) \omega(t) dt. \end{aligned} \quad (60)$$

where $H_m = \max\{\mu_0, \mu_1, \dots, \mu_n\}$, when $t \rightarrow \infty$, it follows

$$\int_0^\infty y^T(t) y(t) dt < \int_0^\infty \gamma^2 \omega^T(t) \omega(t) dt, \quad (61)$$

which means, $\|y(t)\|_2 \leq \gamma \|\omega(t)\|_2$, for $\omega(t) \in l_2[0, \infty)$ with $\gamma = \tilde{\gamma} \sqrt{\frac{\gamma_1}{\gamma_2}}$, $\gamma_1 = \max\{e^{2\delta_1 H_M}, e^{2\delta_2 H_M}\}$, $\gamma_2 = \min\{1, a_2\}$. It is clear that, when $\omega(t) = 0$, the system is asymptotically stable and satisfied the H_∞ performance index γ .

3.2. Fuzzy controller design with mismatched FMF

Theorem 2. For given the sampling period h , the DoS parameters H_M , H_m , positive parameters a_1 and a_2 satisfying $a_1 a_2 > 1$, deception attack probability $\bar{\eta}$, and the constant matrix $\bar{\Phi}$, the system (23) is EMSS and achieves a prescribed H_∞ performance index γ with mismatched FMF, if there exist positive-definite matrices $\bar{P}_i^\zeta$, $\bar{Q}_{\ell i}^\zeta$, $\bar{O}_{\ell i}^\zeta$, MDET weight matrices $\bar{\Psi}_s$, arbitrary matrices $\bar{X}_{\ell i}^\zeta$, $\bar{Y}_{\ell i}$, $\bar{\Omega}_i$, the

controller gains $\bar{K}_{sj}$, and symmetric matrices $\bar{N}_q$ ($q = 1, 2, 3$) with appropriate dimensions, such that the FMF satisfy $g_s(\bar{\theta}(t)) - \varsigma_s g_s(\theta(t)) \geq 0$, and the following inequalities are obtained for any $\dot{g}_k(t) \in [-\mathcal{D}_k, \mathcal{D}_k]$, with $l \neq k$, $k, l, i, s \in \{1, 2, \dots, r\}$, $j \in \{1, 2, \dots, m\}$, $\ell \in \{1, 2\}$, $\iota, \zeta, \mathcal{L} \in \{1, 2\}$:

$$\begin{cases} \bar{P}_l^\zeta + \bar{N}_1 - \bar{P}_k^\zeta \geq 0 \\ \bar{Q}_{1l}^\zeta + \bar{N}_2 - \bar{Q}_{1k}^\zeta \geq 0 \\ \bar{Q}_{2l}^\zeta + \bar{N}_3 - \bar{Q}_{2k}^\zeta \geq 0 \end{cases} \quad (62)$$

$$\dot{g}_k(t) \bar{N}_2 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(\bar{Q}_{1l}^\zeta + \bar{N}_2 - \bar{Q}_{1k}^\zeta) - \bar{X}_1^\zeta \leq 0 \quad (63)$$

$$\dot{g}_k(t) \bar{N}_3 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(\bar{Q}_{2l}^\zeta + \bar{N}_3 - \bar{Q}_{2k}^\zeta) - \bar{X}_2^\zeta \leq 0 \quad (64)$$

$$\begin{cases} \bar{P}_i^1 - a_2 \bar{P}_i^2 \leq 0, & e^{-2(\delta_1 + \delta_2)h} \bar{P}_i^2 - a_1 \bar{P}_i^1 \leq 0 \\ \bar{Q}_{\ell i}^1 - a_2 \bar{Q}_{\ell i}^2 \leq 0, & \bar{X}_1^1 - a_2 \bar{X}_1^2 \leq 0 \\ \bar{Q}_{\ell i}^2 - a_1 \bar{Q}_{\ell i}^1 \leq 0, & \bar{X}_2^1 - a_2 \bar{X}_2^2 \leq 0 \end{cases} \quad (65)$$

$$\begin{bmatrix} \bar{Q}_{2i}^1 & \bar{Z}_1 \\ * & \bar{Q}_{2i}^1 \end{bmatrix} > 0, \quad \begin{bmatrix} \bar{Q}_{2i}^2 & \bar{Z}_2 \\ * & \bar{Q}_{2i}^2 \end{bmatrix} > 0 \quad (66)$$

$$\bar{\Xi}_{is}^\zeta - \bar{\Omega}_i < 0 \quad (67)$$

$$\varsigma_i \bar{\Xi}_{ii}^\zeta + (1 - \varsigma_i) \bar{\Omega}_i < 0 \quad (68)$$

$$\varsigma_s \bar{\Xi}_{is}^\zeta + (1 - \varsigma_s) \bar{\Omega}_i + \varsigma_i \bar{\Xi}_{si}^\zeta + (1 - \varsigma_i) \bar{\Omega}_s < 0, i < s \quad (69)$$

where

$$\bar{\Xi}_{is}^1 = \begin{bmatrix} \bar{\Lambda}_{11}^1 + h \bar{\Gamma}_i^1 & \bar{\Lambda}_{12}^1 \\ * & \bar{\Lambda}_{22}^1 \end{bmatrix} < 0$$

$$\bar{\Xi}_{is}^2 = \begin{bmatrix} \bar{\Lambda}_{11}^2 + h \bar{\Gamma}_i^2 & \bar{\Lambda}_{12}^2 \\ * & \bar{\Lambda}_{22}^2 \end{bmatrix} < 0$$

$$\bar{\Lambda}_{11}^1 = -e^{-2\delta_1 h} \Pi_1^T \bar{O}_1^1 \Pi_1^1 - e^{-2\delta_1 h} \Pi_2^T \bar{O}_2^1 \Pi_2^1 + W_2^T \bar{\Lambda}_3 W_2$$

$$\bar{\Lambda}_{11}^2 = -e^{-2\delta_2 h} \Pi_1^T \bar{O}_1^2 \Pi_1^2 - e^{-2\delta_2 h} \Pi_2^T \bar{O}_2^2 \Pi_2^2 + W_3^T \bar{\Lambda}_4 W_3$$

$$\bar{\Gamma}_i^1 = \text{sym} \left\{ -\frac{1}{h} e^{-2\delta_1 h} \Pi_1^T \bar{Y}_2 \Pi_2^1 \right\} - \frac{1}{h} e^{-2\delta_1 h} \Pi_1^T \bar{O}_1^1 \Pi_1^1 + h W_2^T (\bar{G}_1^T \bar{O}_1^1 \bar{G}_1 + \bar{G}_2^T \bar{O}_1^1 \bar{G}_2) W_2$$

$$\bar{\Gamma}_i^2 = \text{sym} \left\{ -\frac{1}{h} e^{-2\delta_2 h} \Pi_1^T \bar{Y}_2 \Pi_2^2 \right\} - \frac{1}{h} e^{-2\delta_2 h} \Pi_1^T \bar{O}_1^2 \Pi_1^2 + h W_3^T (\bar{G}_1^T \bar{O}_1^2 \bar{G}_1 + \bar{G}_2^T \bar{O}_1^2 \bar{G}_2) W_3$$

$$\bar{\Lambda}_1 = \dot{g}_k(t) \bar{N}_1 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(\bar{P}_l^1 + \bar{N}_1 - \bar{P}_k^1)$$

$$\bar{\Lambda}_2 = \dot{g}_k(t) \bar{N}_1 + \sum_{l=1, l \neq k}^r \mathcal{D}_l(\bar{P}_l^2 + \bar{N}_1 - \bar{P}_k^2)$$

$$\bar{\Lambda}_3 = \begin{bmatrix} \bar{\Lambda}_{111}^1 & \bar{\Lambda}_{112}^1 \\ * & \bar{\Lambda}_{113}^1 \end{bmatrix}, \bar{\Lambda}_{111}^1 = \begin{bmatrix} \bar{\pi}_{11}^1 & \bar{\pi}_{12}^1 & \bar{\pi}_{13}^1 \\ * & \bar{\pi}_{22}^1 & \bar{\pi}_{23}^1 \\ * & * & \bar{\pi}_{33}^1 \end{bmatrix}$$

$$\bar{\Lambda}_{112}^1 = [\bar{\Sigma}_{12} \quad \bar{\Sigma}_{13} \quad \bar{\Sigma}_{14}], \bar{\Lambda}_{113}^1 = \text{diag}\{\bar{\Sigma}_{22}, \bar{\Sigma}_{33}, \bar{\Sigma}_{44}\}$$

$$\bar{N}_{1s} = \sum_{j=1}^m B \bar{K}_{sj}, \bar{N}_{2s} = B[\bar{K}_{s1}, \dots, \bar{K}_{sm}]$$

$$\bar{\pi}_{11}^1 = 2\delta_1 J_1 + \text{sym}(J_1 A_i) + \bar{Q}_{1i}^1 - e^{-2\delta_1 h} \bar{Q}_{2i}^1 + h \bar{X}_1^1 + \bar{\Lambda}_1$$

$$\bar{\pi}_{12}^1 = \bar{\eta} \bar{N}_{1s} + e^{-2\delta_1 h} (\bar{Q}_{2i}^1 + \bar{Z}_1), \bar{\pi}_{13}^1 = -e^{-2\delta_1 h} \bar{Z}_1$$

$$\bar{\pi}_{22}^1 = e^{-2\delta_1 h} (-2\bar{Q}_{2i}^1 - \text{sym}(\bar{Z}_1)) + \lambda_M \bar{\Psi}_s$$

$$\bar{\pi}_{23}^1 = e^{-2\delta_1 h} (\bar{Q}_{2i}^1 + \bar{Z}_1), \bar{\pi}_{33}^1 = -e^{-2\delta_1 h} (\bar{Q}_{1i}^1 + \bar{Q}_{2i}^1)$$

$$\bar{\Sigma}_{12} = \begin{bmatrix} -\bar{\eta} \bar{N}_{2s} \\ -\frac{\lambda_M}{m} \bar{\Psi}_s \\ 0 \end{bmatrix}, \bar{\Sigma}_{13} = \begin{bmatrix} (1 - \bar{\eta}) \bar{N}_{2s} \\ 0 \end{bmatrix}, \bar{\Sigma}_{14} = \begin{bmatrix} D_i \\ 0 \end{bmatrix}$$

$$\bar{\Sigma}_{22} = \text{diag}\left\{ \left(\frac{\lambda_M}{m^2} - \mu_1 \right) \bar{\Psi}_s, \dots, \left(\frac{\lambda_M}{m^2} - \mu_m \right) \bar{\Psi}_s \right\}$$

$$\bar{\Sigma}_{33} = \text{diag}\{-I, \dots, -I\}, \bar{\Sigma}_{44} = -\gamma^2 I$$

$$\bar{\Lambda}_{12}^1 = [\bar{\Lambda}_{121}^1 \quad \bar{\Lambda}_{122}^1], \bar{\Lambda}_{22}^1 = [\bar{\Lambda}_{221}^1 \quad \bar{\Lambda}_{222}^1]$$

$$\bar{\Lambda}_{121}^1 = [J_1 C_i^T \quad J_1 \Phi^T \quad h \bar{G}_1^T \quad h \bar{G}_2^T]$$

$$\bar{\Lambda}_{122}^1 = \begin{bmatrix} \frac{\sqrt{2}h}{2} h \bar{G}_1^T & \frac{\sqrt{2}h}{2} h \bar{G}_2^T & \Pi_1^T \bar{Y}_1 \end{bmatrix}$$

$$\bar{G}_1 = [A_i J_1 \quad \bar{\eta} \bar{N}_{1j} \quad 0 \quad -\bar{\eta} \bar{N}_{2j} \quad (1 - \bar{\eta}) \bar{N}_{2j} \quad D_i]$$

$$\bar{G}_2 = [0 \quad \rho \bar{N}_{1j} \quad 0 \quad -\rho \bar{N}_{2j} \quad -\rho \bar{N}_{2j} \quad 0]$$

$$\bar{\Lambda}_{221}^1 = \text{diag}\{-I, -\bar{\eta} I, e_1^2 \bar{Q}_{2i}^1 - 2e_1 J_1, e_1^2 \bar{Q}_{2i}^1 - 2e_1 J_1\}$$

$$\bar{\Lambda}_{222}^1 = \text{diag}\{e_1^2 \bar{X}_2^1 - 2e_1 J_1, e_1^2 \bar{X}_2^1 - 2e_1 J_1, -e^{-2\delta_1 h} \bar{O}_2^{1'}\}$$

$$\bar{\Lambda}_4 = \begin{bmatrix} \bar{\pi}_{11}^2 & \bar{Q}_{2i}^2 + \bar{Z}_2 & -\bar{Z}_2 & J_2 D_i \\ * & -2\bar{Q}_{2i}^2 - \text{sym}(\bar{Z}_2) & \bar{Q}_{2i}^2 + \bar{Z}_2 & 0 \\ * & * & -e^{2\delta_2 h} \bar{Q}_{1i}^2 - \bar{Q}_{2i}^2 & 0 \\ * & * & * & -\gamma^2 I \end{bmatrix}$$

$$\bar{\pi}_{11}^2 = -2\delta_2 J_2 + \text{sym}(J_2 A_i) + \bar{Q}_{1i}^2 - \bar{Q}_{2i}^2 + h \bar{X}_1^2 + \bar{\Lambda}_2$$

$$\bar{\Lambda}_{12}^2 = [\bar{\Lambda}_{121}^2 \quad \bar{\Lambda}_{122}^2 \quad \bar{\Lambda}_{123}^2 \quad \bar{\Lambda}_{124}^2]$$

$$\bar{\Lambda}_{22}^2 = \text{diag}\{-I, e_2^2 \bar{Q}_{2i}^2 - 2e_2 J_2, e_2^2 \bar{X}_2^2 - 2e_2 J_2, e_2^2 \bar{O}_2^{2'} - 2e_2 J_2\}$$

$$\bar{\Lambda}_{121}^2 = [J_2 C_i^T J_2 \quad 0 \quad 0 \quad 0]^T, \bar{\Lambda}_{124}^2 = [\Pi_2^T \bar{Y}_2]^T,$$

$$\bar{\Lambda}_{122}^2 = [h A_i J_2 \quad 0 \quad 0 \quad h D_i J_2]^T$$

$$\bar{\Lambda}_{123}^2 = \begin{bmatrix} \frac{\sqrt{2}h}{2} h A_i J_2 & 0 & 0 & \frac{\sqrt{2}h}{2} h D_i J_2 \end{bmatrix}^T$$

Proof: Let $J_\varsigma = P_i^{\varsigma-1}$, for $\varsigma = 1, 2$, and define $\Theta_1 =$

$$\text{diag} \left\{ \underbrace{J_1, \dots, J_1}_{3+2m}, I, I, I, \underbrace{J_1, \dots, J_1}_5 \right\}. \text{ Multiplying both sides}$$

of Ξ_{is}^1 by Θ_1 and Θ_1^T yields the transformed inequality.

Furthermore, define $\bar{K}_{sj} = K_{sj} J_1^T$. Introduce also $\bar{Q}_{\ell i}^1 = J_1 \bar{Q}_{\ell i}^1 J_1^T$, $\bar{O}_i^1 = J_1 \bar{O}_i^1 J_1^T$, $\bar{S}_{\ell i}^1 = J_1 \bar{S}_{\ell i}^1 J_1^T$, $\bar{X}_i^1 = J_1 \bar{X}_i^1 J_1^T$, and $\bar{\Psi}_s = J_1 \bar{\Psi}_s J_1^T$. For the inequality, the non-linear term $-J_1 \bar{Q}_{2i}^1 J_1$ can be replaced by $e_1^2 \bar{Q}_{2i}^1 - 2e_1 J_1$ due to $(\bar{Q}_{2i}^1 - e_1^{-1} P_i^1)(\bar{Q}_{2i}^1)^{-1}(\bar{Q}_{2i}^1 - e_1^{-1} P_i^1) \geq 0$, and similar results can be obtained for $\bar{X}_2^1$ and $\bar{O}_2^{1'}$.

On the other hand, let $\Theta_2 = \text{diag}\{J_2, J_2, J_2, I, I, J_2, J_2, J_2\}$. Multiplication of Θ_2 and Θ_2^T on the two sides of Ξ_{is}^2 is performed, respectively. Furthermore, denote $\bar{Q}_{\ell i}^2 =$

$J_2 Q_{\ell}^2 J_2^T$, $\bar{O}_{\ell}^2 = J_2 O_{\ell}^2 J_2^T$, $\bar{X}_q^2 = J_2 X_q^2 J_2^T$, and $\bar{S}_{\ell i}^2 = J_2 S_{\ell i}^2 J_2^T$. Similarly, the non-linear term $-J_2 O_{2i}^2 J_2$ can be replaced by $e_2^2 \bar{O}_{2i}^2 - 2e_2 J_2$ owing to $(\bar{O}_{2i}^2 - e_2^{-1} P_i^2)(\bar{O}_{2i}^2)^{-1}(\bar{O}_{2i}^2 - e_2^{-1} P_i^2) \geq 0$.

To address the mismatched FMF, slack matrix $\bar{\Omega}_i$ is introduced satisfying the following zero equality:

$$\sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\tilde{\theta}(t)) \bar{\Omega}_i = \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\theta(t)) \bar{\Omega}_i. \quad (70)$$

Moreover, utilizing the predefined boundary condition $g_s(\tilde{\theta}(t)) - \varsigma_s g_s(\theta(t)) \geq 0$, one obtained:

$$\begin{aligned} & \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\tilde{\theta}(t)) \bar{\Xi}_{is}^{\zeta} \\ &= \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) (g_s(\tilde{\theta}(t)) - \varsigma_s g_s(\theta(t))) (\bar{\Xi}_{is}^{\zeta} - \bar{\Omega}_i) \\ &+ \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\theta(t)) (\varsigma_s \bar{\Xi}_{is}^{\zeta} - \varsigma_s \bar{\Omega}_i + \bar{\Omega}_i). \end{aligned} \quad (71)$$

Subsequently, the completely matched summation part is reorganized into its diagonal and cross components:

$$\begin{aligned} & \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) g_s(\tilde{\theta}(t)) \bar{\Xi}_{is}^{\zeta} \\ &= \sum_{i=1}^r \sum_{s=1}^r g_i(\theta(t)) (g_s(\tilde{\theta}(t)) - \varsigma_s g_s(\theta(t))) (\bar{\Xi}_{is}^{\zeta} - \bar{\Omega}_i) \\ &+ \sum_{i=1}^r g_i^2(\theta(t)) (\varsigma_i \bar{\Xi}_{ii}^{\zeta} - \varsigma_i \bar{\Omega}_i + \bar{\Omega}_i) + \sum_{i=1}^{r-1} \sum_{s=i+1}^r \\ &g_i(\theta(t)) g_s(\theta(t)) (\varsigma_s \bar{\Xi}_{is}^{\zeta} - \varsigma_s \bar{\Omega}_i + \bar{\Omega}_i + \varsigma_i \bar{\Xi}_{si}^{\zeta} - \varsigma_i \bar{\Omega}_s + \bar{\Omega}_s). \end{aligned} \quad (72)$$

Based on the above analysis, the linear matrix inequalities in this theorem guarantee that $E\{\dot{V}_1(t)\} < 0$ and $E\{\dot{V}_2(t)\} \leq 0$. As a result, system (23) is EMSS with the H_{∞} performance. Additionally, the controller gains are given by $K_{sj} = \bar{K}_{sj} (J_1^T)^{-1}$.

4. Simulation

In this section, three simulation cases are discussed to verify the feasibility and superiority of the control strategy proposed in this paper:

First, given the matrices A_{η} , A_v and B_{uw} in (4) as:

$$\begin{aligned} A_{\eta} &= -M^{-1}G = \begin{bmatrix} -0.0358 & 0 & 0 \\ 0 & -0.0208 & 0 \\ 0 & -0.0394 & 0 \end{bmatrix}, \\ A_v &= -M^{-1}N = \begin{bmatrix} -0.0797 & 0 & 0 \\ 0 & -0.0818 & -0.0577 \\ 0 & -0.2254 & -0.2535 \end{bmatrix}, \end{aligned}$$

$$B_{uw} = M^{-1} = \begin{bmatrix} 0.9215 & 0 & 0 \\ 0 & 0.7802 & 1.4811 \\ 0 & 1.4811 & 7.4562 \end{bmatrix}.$$

Accordingly, for $\theta(t) \in [-\frac{\pi}{6}, \frac{\pi}{6}]$, the nonlinear USV can be modeled by the subsequent local T-S fuzzy systems:

$$\begin{aligned} A_i &= \begin{bmatrix} 0 & J_i \\ A_{\eta} & A_v \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ B_{uw} \end{bmatrix}, \\ C_{2i} &= [0 \quad 0 \quad -0.002 \quad 0 \quad 0 \quad 0.01], i \in \{1, 2, 3, 4\} \end{aligned}$$

where the corresponding rotational matrices are defined as:

$$\begin{aligned} J_1 &= \begin{bmatrix} 1 & -1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, J_2 = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ J_3 &= \begin{bmatrix} 1 & 1/2 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, J_4 = \begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 \\ -1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Furthermore, the FMF are explicitly constructed as:

$$\begin{aligned} g_1(\theta(t)) &= \phi_1(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_2(\theta(t)) &= \phi_1(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_3(\theta(t)) &= \phi_2(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_4(\theta(t)) &= \phi_2(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \end{aligned}$$

in which the intermediate variables ϕ_1 and ϕ_2 are given by:

$$\phi_1(x_3(t)) = \frac{1}{2} + \sin(x_3(t)), \quad \phi_2(x_3(t)) = \frac{1}{2} - \sin(x_3(t)).$$

The simulation essential parameters are configured as follows: $\lambda_M = 0.5$, $\lambda_m = 0.01$, $\wp = 100$, $h = 0.2$ s, $\dot{g}_k(t) \in [-0.5, 0.5]$. The exponential decay rates are chosen as $\delta_1 = 0.08$ and $\delta_2 = 0.65$, and $e_1 = 0.2$, $e_2 = 0.2$, $a_1 = 1.15$, $a_2 = 1.15$, and $\gamma = 20$, the mismatched membership bounds are defined as $\varsigma_i = 0.85$ ($i \in \{1, 2, 3, 4\}$). Meanwhile, the time-varying environmental disturbances $w(t)$ are described as:

$$\begin{cases} w_1(t) = -0.1e^{-t} \\ w_2(t) = -0.4 \cos(1.58t)e^{-0.235t} \\ w_3(t) = -0.2 \sin(2t)e^{-2t} \end{cases} \quad (73)$$

Case 1. State responses and control effectiveness verification under hybrid cyber-attacks.

The key attack parameters are set as $\bar{\eta} = 0.80$, $\mathcal{H}_M = 2.10$ s, and $\mathcal{H}_m = 0.20$ s. The deception attack is modeled as $f(x(t)) = -\tanh(Lx(t))$ with $L = \text{diag}\{0.15, 0.05, 0.2, 0.15, 0.15, 0.2\}$, which satisfies the bound $\Phi = 0.2I_{6 \times 6}$. The proposed MDETMs sets $m = 3$ with weights $\mu_1 = 0.6$, $\mu_2 = 0.3$, and $\mu_3 = 0.1$. By solving the conditions in Theorem ??, the feedback control gains alongside the triggering weight matrices are computed as follows:

$$K_{11} = \begin{bmatrix} -1.1990 & -0.1299 & -0.0041 & -1.7782 & -0.0696 & -0.0069 \\ 0.2418 & -2.2834 & 0.3883 & 0.1289 & -3.4120 & 0.5784 \\ -0.0469 & 0.4520 & -0.1862 & -0.0252 & 0.6895 & -0.2595 \end{bmatrix},$$

$$\begin{aligned}
 K_{12} &= \begin{bmatrix} -0.7055 & -0.0745 & -0.0024 & -1.0478 & -0.0397 & -0.0041 \\ 0.1387 & -1.3435 & 0.2293 & 0.0735 & -2.0107 & 0.3418 \\ -0.0269 & 0.2660 & -0.1097 & -0.0144 & 0.4064 & -0.1530 \end{bmatrix}, \\
 K_{13} &= \begin{bmatrix} -0.3659 & -0.0373 & -0.0013 & -0.5444 & -0.0197 & -0.0022 \\ 0.0694 & -0.6970 & 0.1193 & 0.0364 & -1.0447 & 0.1780 \\ -0.0135 & 0.1380 & -0.0569 & -0.0071 & 0.2112 & -0.0796 \end{bmatrix}, \\
 K_{21} &= \begin{bmatrix} -1.2142 & -0.1581 & -0.0065 & -1.8267 & -0.0845 & -0.0100 \\ 0.2942 & -2.3114 & 0.3894 & 0.1573 & -3.5013 & 0.5824 \\ -0.0573 & 0.4575 & -0.1865 & -0.0312 & 0.7070 & -0.2603 \end{bmatrix}, \\
 K_{22} &= \begin{bmatrix} -0.7143 & -0.0905 & -0.0039 & -1.0759 & -0.0481 & -0.0059 \\ 0.1683 & -1.3599 & 0.2298 & 0.0895 & -2.0624 & 0.3440 \\ -0.0328 & 0.2693 & -0.1098 & -0.0177 & 0.4166 & -0.1535 \end{bmatrix}, \\
 K_{23} &= \begin{bmatrix} -0.3706 & -0.0458 & -0.0021 & -0.5590 & -0.0242 & -0.0031 \\ 0.0852 & -0.7056 & 0.1194 & 0.0450 & -1.0716 & 0.1790 \\ -0.0166 & 0.1398 & -0.0570 & -0.0089 & 0.2165 & -0.0798 \end{bmatrix}, \\
 K_{31} &= \begin{bmatrix} -1.1990 & 0.1299 & 0.0041 & -1.7782 & 0.0696 & 0.0069 \\ -0.2418 & -2.2834 & 0.3883 & -0.1289 & -3.4120 & 0.5784 \\ 0.0469 & 0.4520 & -0.1862 & 0.0252 & 0.6895 & -0.2595 \end{bmatrix}, \\
 K_{32} &= \begin{bmatrix} -0.7055 & 0.0745 & 0.0024 & -1.0478 & 0.0397 & 0.0041 \\ -0.1387 & -1.3435 & 0.2293 & -0.0735 & -2.0107 & 0.3418 \\ 0.0269 & 0.2660 & -0.1097 & 0.0144 & 0.4064 & -0.1530 \end{bmatrix}, \\
 K_{33} &= \begin{bmatrix} -0.3659 & 0.0373 & 0.0013 & -0.5444 & 0.0197 & 0.0022 \\ -0.0694 & -0.6970 & 0.1193 & -0.0364 & -1.0447 & 0.1780 \\ 0.0135 & 0.1380 & -0.0569 & 0.0071 & 0.2112 & -0.0796 \end{bmatrix}, \\
 K_{41} &= \begin{bmatrix} -1.2142 & 0.1581 & 0.0065 & -1.8267 & 0.0845 & 0.0100 \\ -0.2942 & -2.3114 & 0.3894 & -0.1573 & -3.5013 & 0.5824 \\ 0.0573 & 0.4575 & -0.1865 & 0.0312 & 0.7070 & -0.2603 \end{bmatrix}, \\
 K_{42} &= \begin{bmatrix} -0.7143 & 0.0905 & 0.0039 & -1.0759 & 0.0481 & 0.0059 \\ -0.1683 & -1.3599 & 0.2298 & -0.0895 & -2.0624 & 0.3440 \\ 0.0328 & 0.2693 & -0.1098 & 0.0177 & 0.4166 & -0.1535 \end{bmatrix}, \\
 K_{43} &= \begin{bmatrix} -0.3706 & 0.0458 & 0.0021 & -0.5590 & 0.0242 & 0.0031 \\ -0.0852 & -0.7056 & 0.1194 & -0.0450 & -1.0716 & 0.1790 \\ 0.0166 & 0.1398 & -0.0570 & 0.0089 & 0.2165 & -0.0798 \end{bmatrix}, \\
 \Psi_1 &= \begin{bmatrix} 0.3808 & -0.0003 & -0.0009 & 0.5368 & -0.0004 & -0.0005 \\ -0.0003 & 0.3846 & -0.0067 & 0.0001 & 0.5400 & -0.0173 \\ -0.0009 & -0.0067 & 0.1932 & -0.0002 & -0.0319 & 0.2455 \\ 0.5368 & 0.0001 & -0.0002 & 0.8082 & -0.0000 & 0.0001 \\ -0.0004 & 0.5400 & -0.0319 & -0.0000 & 0.8120 & -0.0553 \\ -0.0005 & -0.0173 & 0.2455 & 0.0001 & -0.0553 & 0.3342 \end{bmatrix}, \\
 \Psi_2 &= \begin{bmatrix} 0.3812 & -0.0004 & -0.0006 & 0.5543 & -0.0019 & -0.0000 \\ -0.0004 & 0.3847 & -0.0071 & 0.0016 & 0.5570 & -0.0184 \\ -0.0006 & -0.0071 & 0.1932 & 0.0005 & -0.0320 & 0.2456 \\ 0.5543 & 0.0016 & 0.0005 & 0.8429 & -0.0001 & 0.0010 \\ -0.0019 & 0.5570 & -0.0320 & -0.0001 & 0.8459 & -0.0562 \\ -0.0000 & -0.0184 & 0.2456 & 0.0010 & -0.0562 & 0.3343 \end{bmatrix}, \\
 \Psi_3 &= \begin{bmatrix} 0.3808 & 0.0003 & 0.0009 & 0.5368 & 0.0004 & 0.0005 \\ 0.0003 & 0.3846 & -0.0067 & -0.0001 & 0.5400 & -0.0173 \\ 0.0009 & -0.0067 & 0.1932 & 0.0002 & -0.0319 & 0.2455 \\ 0.5368 & -0.0001 & 0.0002 & 0.8082 & 0.0000 & -0.0001 \\ 0.0004 & 0.5400 & -0.0319 & 0.0000 & 0.8120 & -0.0553 \\ 0.0005 & -0.0173 & 0.2455 & -0.0001 & -0.0553 & 0.3342 \end{bmatrix}, \\
 \Psi_4 &= \begin{bmatrix} 0.3812 & 0.0004 & 0.0006 & 0.5543 & 0.0019 & 0.0000 \\ 0.0004 & 0.3847 & -0.0071 & -0.0016 & 0.5570 & -0.0184 \\ 0.0006 & -0.0071 & 0.1932 & -0.0005 & -0.0320 & 0.2456 \\ 0.5543 & -0.0016 & -0.0005 & 0.8429 & 0.0001 & -0.0010 \\ 0.0019 & 0.5570 & -0.0320 & 0.0001 & 0.8459 & -0.0562 \\ 0.0000 & -0.0184 & 0.2456 & -0.0010 & -0.0562 & 0.3343 \end{bmatrix}.
 \end{aligned}$$

Assuming the initial USV state $x(0) = [0.1, -0.1, 0.2, -0.2, 0.3, -0.3]^T$ and a simulation time of $T = 20$ s, the simulation results of the closed-loop USV system are illustrated in Figs. 3-8. Fig. 3 presents the state response trajectories of the system. It is evident that all state variables converge rapidly and smoothly to zero despite the continuous interference from deception attacks, DoS attacks, and external marine disturbances. This demonstrates the strong robustness of the proposed control scheme. Fig. 4 plots the corresponding control input signals $u(t)$, which eventually stabilize at zero as the USV reaches the equilibrium point.

Furthermore, Figs. 5 and 6 depict the triggering instants for the four fuzzy subsystems and the trajectories of the adaptive threshold parameters $\lambda_i(t)$, respectively. As shown in Fig. 6, the triggering thresholds of the proposed MDETM are dynamically adjusted based on the system state errors and the cyber-attack intensity, thereby effectively mitigating the attack impacts while strictly conserving communication resources. Specifically, as illustrated in Fig. 5, fuzzy subsystems 1-4 are triggered 63, 59, 63, and 59 times, respectively. Fig. 7 illustrates the occurring instants of deception and DoS attacks. Fig. 8 illustrates the variation results of the FMF and their derivatives for each fuzzy subsystem, the derivatives of the FMF satisfy the predefined conditions.

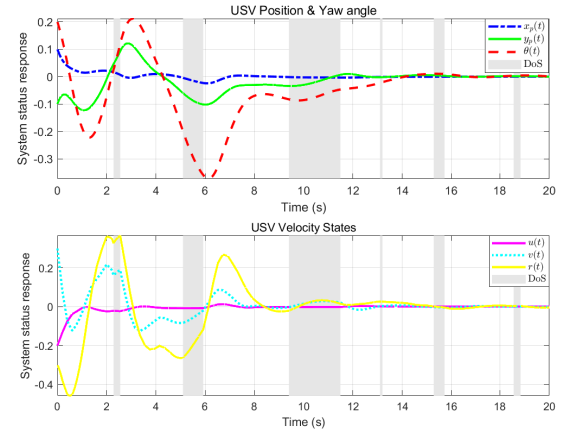


Figure 3: State response trajectories with MDETM.

Case 2. Parameter analysis and communication resource optimization of the proposed MDETM

To further explore the pivotal role of historical data and threshold bounds in the proposed MDETM, the sampling period is set to $h = 0.02$ s, and the weights are chosen as $\mu_1 = 0.90$, $\mu_2 = 0.05$, and $\mu_3 = 0.05$, while the remaining parameters are identical to those in Case 1. Table 1 presents the average data transmission rate (ADTR) under various memory depths m , weighting coefficients μ_i , and maximum adaptive thresholds λ_M . As illustrated in Table 1, the maximum adaptive threshold λ_M exhibits a negative correlation with the ADTR. Notably, an excessively large threshold yields limited resource optimization and may inversely force frequent control updates to maintain stability.

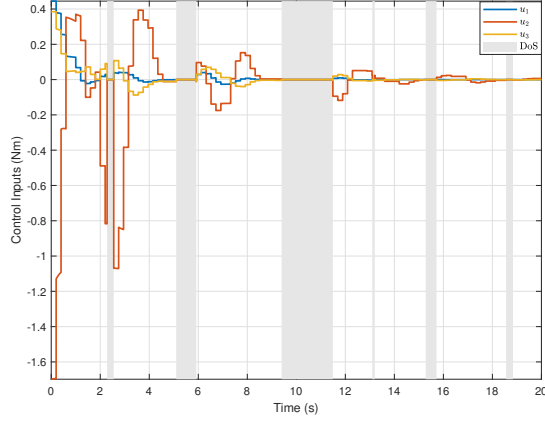
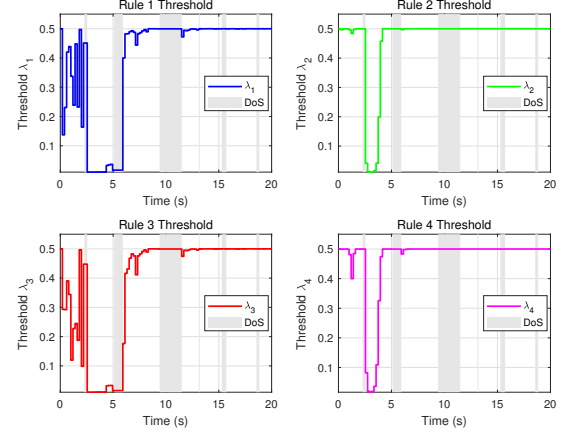
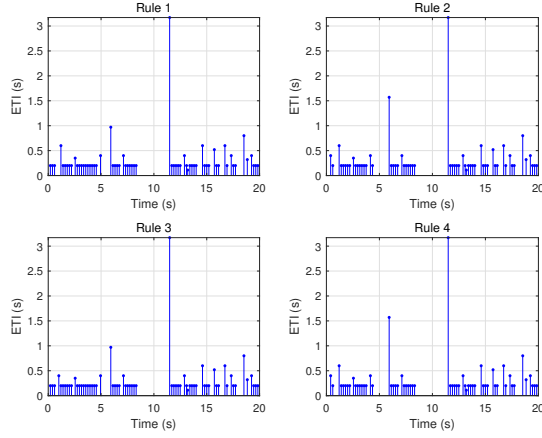

 Figure 4: Trajectories of the control input $u(t)$.

 Figure 6: Evolution of the adaptive threshold $\lambda_i(t)$.


Figure 5: Triggering instants and release intervals.

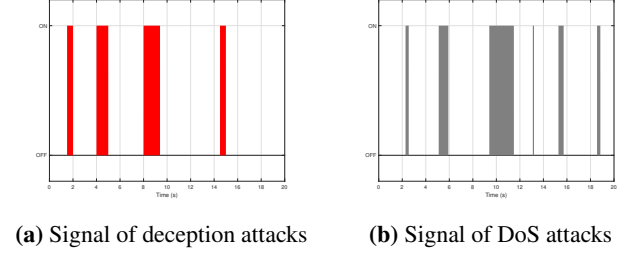


Figure 7: Situation of hybrid cyber-attacks in Case 1.

 Table 1
ADTR under varying MDET parameters

ADTR λ_M	m, μ_i	$m = 1$	$m = 3$		$m = 4$
		$\mu_i = 1$	$\{0.5, 0.25, 0.25\}$	$\{0.8, 0.1, 0.1\}$	$\{0.25 \dots 0.25\}$
0.1		10.25%	22.00%	16.83%	31.60%
0.3		6.25%	13.60%	10.57%	20.40%
0.5		5.25%	11.03%	8.30%	16.40%
0.7		4.44%	9.63%	6.68%	13.80%
1.0		3.65%	8.13%	4.98%	12.50%

Thus, a dynamic balance between control performance and resource conservation is indispensable. Furthermore, although the memoryless mechanism $m = 1$ yields a numerically lower ADTR, this reduction is fundamentally achieved at the severe expense of system control performance. Particularly in scenarios involving hybrid cyber-attacks and external disturbances, introducing historical state information and judiciously configuring historical weight allocations are of profound practical significance for enhancing the overall robustness of the system.

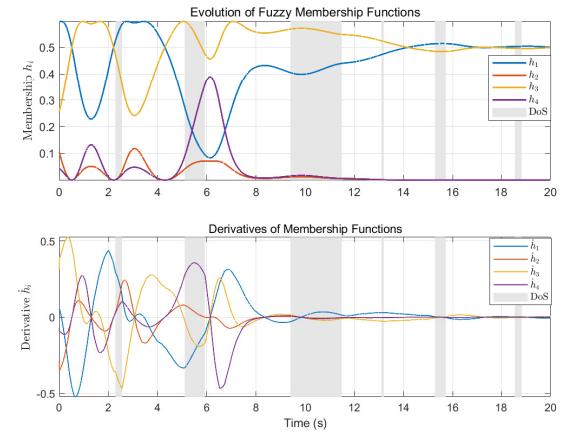


Figure 8: Fuzzy membership function and its derivatives.

Furthermore, to evaluate the resource-saving efficacy and control performance, comparative metrics including the maximum triggering threshold (MTT), triggering counts (TCs), average triggering interval (ATI), the integral of absolute error (IAE) under various control strategies (VCSs) are tabulated in Table 2. Existing control schemes, such as adaptive memory ETM (AMETM, Xu et al. (2025)), attack-related (ARETM, Lin et al. (2025)), and dynamic ETM (DETM, Ma et al. (2025)), implement various mechanisms

Table 2

Performance comparisons of different communication schemes

Scheme	MTT	TCs	ATI (s)	IAE
Periodic Sampled	—	802	0.025	5.9077
DMETM Liu et al. (2025)	0.50	132	0.152	5.1365
AMETM Xu et al. (2025)	0.50	105	0.190	4.8826
DETM Ma et al. (2025)	0.50	110	0.182	5.0947
ARETM Lin et al. (2025)	0.50	61	0.328	4.9504
Fuzzy-independent MDETMM	0.50	77	0.260	5.0055
Memoryless MDETMM	0.50	47, 39, 49, 42	0.452	5.8668
Proposed MDETMM	0.50	57, 50, 60, 50	0.369	4.8168

to mitigate system errors and cyberattacks. Compared to these methods, the proposed method achieves a superior improvement in control tracking while maximizing communication efficiency. Specifically, the proposed MDETMM prolongs the ATI to 0.369s while maintaining a low IAE of 4.8168. Although the memoryless MDETMM yields the lowest TCs, its control performance severely degrades, with the IAE spiking to 5.8668. Ultimately, the proposed algorithm demonstrates distinct superiority in achieving an optimal balance between communication alleviation and resilient control under hybrid cyber-attack environments.

Case 3. Stability comparison under varying attack intensities

To thoroughly evaluate the resilience and superiority of the proposed MDETMM, 3 compares the system state responses under four distinct cyber-attack scenarios (Fig. 9), where $\bar{\alpha}$ and $\bar{\eta}$ denote the expected probabilities of DoS and deception attacks, respectively. As shown in Fig. 9a and 9b, under mild attacks, the controller swiftly compensates for missing packets and ensures rapid state convergence. When attack intensities escalate to 50% and 60% (Fig. 9c and 9d), despite prolonged network blockages and frequent malicious injections, the MDETMM effectively neutralizes these severe threats via historical data compensation and dynamically adjusted thresholds. Ultimately, all system states successfully converge to the equilibrium point, explicitly demonstrating the exceptional robustness of the proposed approach in hostile environments.

To isolate the effects of random deception attacks and validate the superiority of the proposed method, Table 3 records the IAE of various schemes under different DoS attack probabilities $\bar{\alpha}$. As the attack intensity increases, the IAE values for all methods inevitably rise, objectively reflecting the adverse impact of prolonged data dropouts on system tracking accuracy. Notably, the methods proposed by Xu et al. (2025) sequentially lose their efficacy as the attack severity escalates. Furthermore, compared to the Memoryless MDETMM and the Looped-independent approach, the proposed MDETMM successfully preserves outstanding control performance even under extreme communication blockages $\bar{\alpha} = 0.7$. This explicitly demonstrates that the proposed dynamic thresholding and memory compensation strategies not only endow the system with exceptional robustness during consecutive packet losses but also exhibit significant superiority in reducing design conservatism.

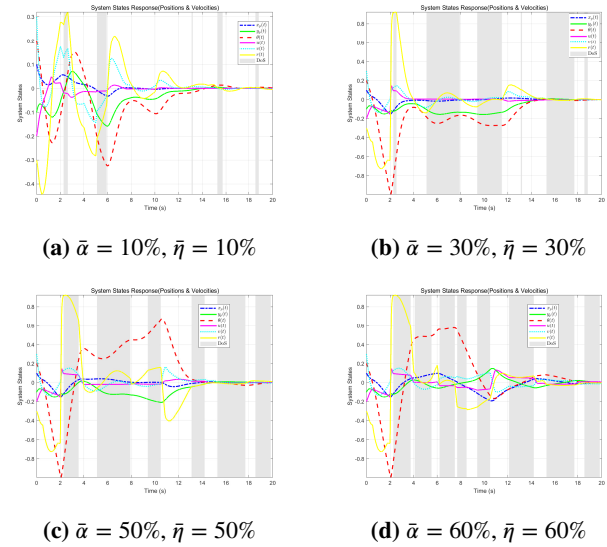

Figure 9: System state response under varying attack intensities

Table 3

 IAE performance under different $\bar{\alpha}$ for different methods

IAE	$\bar{\alpha}$					
	Method	0	0.2	0.4	0.6	0.7
	Xu et al. (2025)	4.83	5.05	5.59	7.05	—
	Memoryless MDETMM	4.12	4.15	5.05	7.17	14.22
	Looped-independent MDETMM	3.13	3.76	5.01	6.02	9.97
	Proposed MDETMM	2.71	3.49	4.71	5.80	9.88

5. Conclusion

This article presents a novel MDETMM for T–S fuzzy USV systems under hybrid cyber-attacks. First, a unified T–S fuzzy USV model is established by integrating ETM, network delays, FMF, and hybrid cyber-attacks. Second, the limitations of conventional ETM in resisting attacks and conserving communication resources are analyzed, based on which an attack-related MDETMM is designed to enhance cyber-attack resilience and improve communication efficiency. Third, a novel LKF is constructed based on the TSLFM to effectively reduce design conservatism, from which sufficient conditions are derived to guarantee the EMSS of the switched system and the H_∞ performance. Finally, the simulation results verify the effectiveness of the proposed approach, and further comparative studies demonstrate the superiority of the proposed control method. Future research will focus on more complex fuzzy USV scenarios, such as imperfect premise matching and hybrid triggering mechanisms, and also aim to reduce the computational complexity of the proposed method.

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